

From gate to runway: A systematic review of airport ground operations optimization

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ABSTRACT

Over the past few decades, the domain of airline and airport ground operations has been extensively studied by numerous researchers. Many of these studies have focused on optimizing various aspects of airline and airport ground operations using operations research techniques. We provide an in-depth review of the literature in this domain comprising 199 scientific papers. We have employed bibliometric and network analysis tools to complete a systematic literature mapping, identifying 7 key functional areas in airport ground operations: airport gate assignment, boarding process and strategies, turnaround process, baggage handling, ground workforce planning and scheduling, aircraft taxiing and ground movements, and ground support equipment handling. For each area, we explore problems, modeling approaches, and solution methods. Furthermore, the review identifies specific future research directions relevant to the 7 key functional areas of ground operations. The overarching prospective research paths center on algorithmic advancements, adaptability to uncertainty, integration between subdomains, and inclusion of external aspects.

1. Introduction

In recent decades, global commercial air traffic has exhibited substantial growth, with aircraft movements increasing by an average of 3.2% annually between 2010 and 2019 (ACI, 2010, 2019). Although the COVID-19 pandemic abruptly disrupted this upward trajectory, clear signs of recovery are now emerging, with air travel demand rebounding at a faster pace than initially projected (IATA, 2019b, 2024). The substantial increase in global air traffic over recent years has resulted in heavily congested airport infrastructure and a notable rise in flight delays. Operating under high capital costs and low operating margins, airlines continuously pursue time and cost reductions to remain competitive (Cavada et al., 2023). Disruptions affecting passengers' connections, crew availability, and ground operations are major sources of cost for airlines (Cook and Tanner, 2011). To maximize the revenue from an aircraft, airlines aim to minimize ground time, resulting in tight turnaround windows for ground operations tasks. According to Oreschko et al. (2012), one of the main causes of flight delays is poor planning of ground operations. In particular, disruptions in ground operations, such as early or late flight arrivals and task duration deviations, can lead to delay propagation, where subsequent tasks assigned to employees are also delayed (Poullet and Parmentier, 2020). The impact of these delays is well-documented; for instance, IATA (2019a) reported that the annual cost of flight delays to

the global economy was estimated at \$50 billion in 2019, underscoring the critical need for effective ground operations management.

The resurgence of air travel has revived the challenges associated with airport or airline ground operations, emphasizing the importance of efficiently optimizing these processes. Collaborative coordination among all stakeholders involved, including airports, airlines, air traffic management, and ground handlers, as well as improving the planning of airport resources, are essential for enhancing the operational efficiency of air transportation. Many aircraft delays can be attributed to prolonged turnaround times resulting from a lack of integrated planning across diverse ground activities and inefficient resource utilization. Furthermore, ground tasks are highly interdependent, where each operation has the potential to generate delays that can easily propagate to other ground operations and broader airport processes (Fricke and Schultz, 2009). Historically, airport or airline internal divisions have conducted these ground operations. The deregulation of the ground operations market, especially in Europe, has led to a significant increase in the number of third-party service providers (Salerno et al., 2012). This new landscape, characterized by multiple ground handlers offering a range of services, further emphasizes the critical importance of efficiently scheduling ground operations.

The field of airline and airport operations was still relatively immature in 2003, as noted by Barnhart et al. (2003). Since then, the

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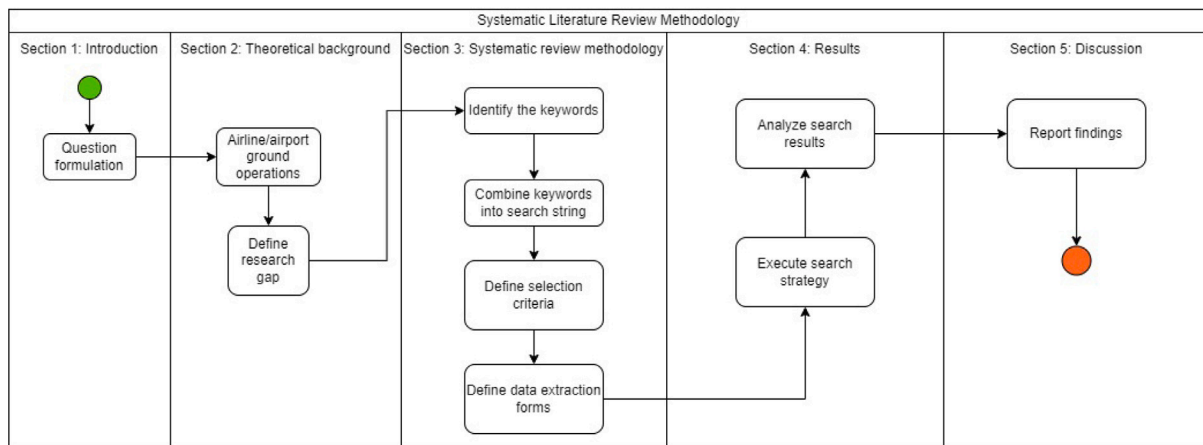


Fig. 1. Visualization of the systematic review approach. All activities are classified into five sections as proposed by Denyer and Tranfield (2009).

field has undergone significant advancement in multiple directions, largely fueled by the rapid growth in computing power. This growth has enabled more sophisticated modeling of complex decision problems and the development of advanced solution methods that can still be solved in a reasonable time. This literature review aims to systematically structure and analyze the extensive body of research that has emerged from these advancements, providing a comprehensive overview of the current state of the field. In this systematic literature review (SLR), we seek to provide a comprehensive overview of the existing literature on optimization problems for airport ground operations using decision-making techniques or operations research techniques. Although the literature on this topic has expanded drastically over the last few decades, a structured overview is lacking. We analyze and synthesize the current literature, identify optimization methods used, and explore open future research directions. This SLR answers the following three research questions:

1. RQ1: What are the key functional areas in airline or airport ground operations that focus on optimization of processes?
2. RQ2: What are the types of optimization techniques and solution methods used in optimizing ground operations?
3. RQ3: What are the future research directions per identified functional area in airline ground operations?

We use the systematic review methodology proposed by Denyer and Tranfield (2009) to answer these research questions. Fig. 1 provides a visualization of the systematic review approach, which classifies all activities into 5 sections that form the structure of this paper. Section 2 provides the theoretical background and defines the research gap that this SLR addresses. In Section 3, we explain the methodology used to conduct the SLR. The search results, bibliometric analysis and the results for RQ1 are explained in Section 4. Section 5 discusses the identified key functional areas and answers RQ2 and RQ3. Section 6 concludes this paper.

2. Theoretical background and contribution

Ground operations procedures can be divided into two types: terminal and ramp (Padrón et al., 2016). Terminal activities are performed inside the terminal building and concern passenger services. Ramp operations take place when the aircraft is parked, between the time it arrives at the stand and its departure (Padrón et al., 2016). Taxiing and ground movement occur before the aircraft reaches the stand upon arrival and after it leaves the stand for departure. A crucial aspect of ramp operations is the timely assignment of a gate immediately after landing to facilitate an efficient passenger disembarkation process. Boarding can follow various strategies, depending on airline preferences and

operational constraints. Optimizing the ground movements of aircraft, such as taxiing routes, can significantly reduce delays and improve overall airport efficiency. Fig. 2 shows the key ground operations performed during a typical aircraft turnaround.

The aircraft turnaround process is highly complex, with its duration depending on numerous operational factors. These include factors such as the aircraft type, the number of ground operations tasks, the parking position (either at a contact or remote stand), the required service time to complete each task, and various precedence constraints. The turnaround process is concluded at the off-block time, characterized by the closure of all aircraft doors, the removal of the boarding bridge, the presence of the pushback vehicle, and the readiness of the aircraft for startup and pushback. While this pushback operation may not be necessary for aircraft parked at remote stands, it remains the most conventional method for exiting the parking position. All these tasks must be accomplished with the support of the ground-handling workforce and the equipment.

The proper scheduling and planning of staff and equipment plays an essential role in ground operations. Existing research has highlighted the interconnected nature of airport ground operations, where scheduling decisions for one service can impact other related activities Gök et al. (2020). Tasks associated with the same aircraft are subject to precedence constraints and time window considerations, making isolated optimization ineffective at the system level. This interdependence is further emphasized by Atkin et al. (2010), who consider the airport ground movement problem as a large-scale routing and scheduling task integrating arrival and departure sequencing, and gate assignment. Their study demonstrates that optimizing individual sub-processes in isolation is insufficient and explicitly calls for integrated and robust optimization strategies to address the dynamic and uncertain nature of real airport operations. Consequently, a significant focus in the literature has been on developing optimization models to manage airport ground operations (Deng et al., 2024; Guimarães and Padrón, 2022; Guo et al., 2020). Recently, Cappanera et al. (2024) propose an integrated mixed integer linear programming (MILP) framework for joint task scheduling and personnel rostering of airport ground staff, showing that simultaneous optimization significantly outperforms sequential planning in large-scale real-world airport settings. In a complementary direction, Zhang et al. (2025) propose a prescriptive analytics framework for airport gate assignment that combines machine learning based arrival time prediction with two-stage stochastic programming to explicitly account for operational uncertainty. Their results show that estimate-then-optimize strategies provide more robust and reliable gate plans than predict-then-optimize approaches.

The literature review by Barnhart et al. (2003) on the application of operations research in the air transport industry focused on specific

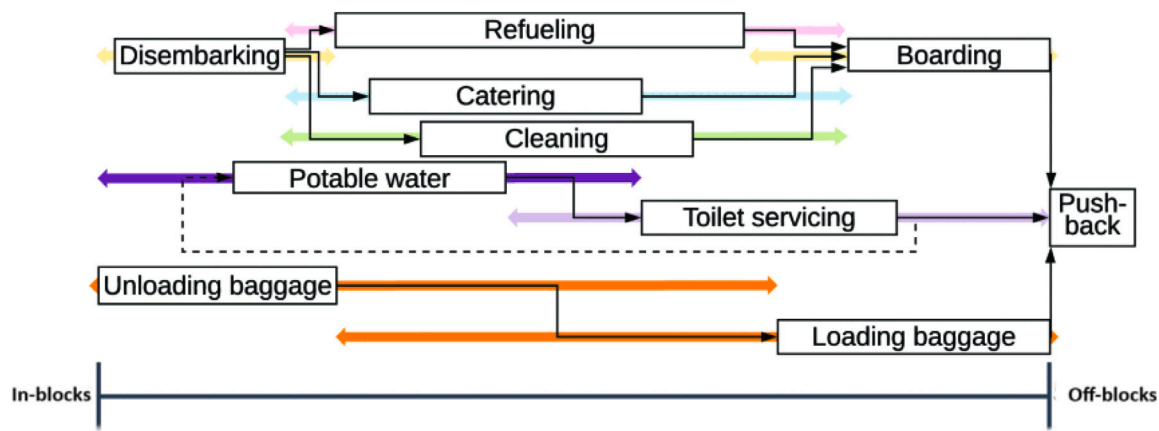


Fig. 2. Activity flow of a turnaround of an aircraft.
Source: Gök et al. (2020).

domains, including aircraft and crew schedule planning, revenue management, as well as the planning and operations of airports and air traffic management. At the time, the field was still relatively immature. However, substantial progress has since been made, leading to several reviews on topics such as airline revenue management (Raza et al., 2020), airline scheduling optimization and planning (Xu et al., 2024; Barnhart, 2008; Eltoukhy et al., 2017), airline disruption management (Hassan et al., 2021), and crew scheduling (Deveci and Demirel, 2018). In the specific area of ground operations, several review studies have been conducted on subtopics such as airport gate assignment (Daş et al., 2020), boarding processes (Qiang et al., 2022; Bidanda et al., 2017), the turnaround process (Schmidt, 2017), aircraft taxiing (Lukic et al., 2018), and ground workforce planning and scheduling (Wandelt and Wang, 2024).

While several studies have focused on specific aspects of ground operations, a comprehensive review of the field as a whole is lacking. Given the interconnected nature of ground operations, where multiple processes rely on shared resources and central data storage, an integrated review is essential. The availability of granular data further enables a more holistic understanding of operational challenges and optimization opportunities. Moreover, considering the high practical relevance of efficient ground operations for airlines and airports, a broad review can provide valuable insights that extend beyond the scope of focused studies. This paper aims to bridge this gap by providing a comprehensive analysis of the key research areas in this domain, highlighting synergies between different approaches and identifying opportunities for future research in optimizing ground operations. Therefore, following the overarching review of operations research in all aspects of the air transport industry, we focus on summarizing and classifying literature on airport ground operations from 2004 to 2024.

3. Systematic review methodology

In this Section 3, we explain how we conduct the SLR by defining the search strategy (Section 3.1), selection criteria (Section 3.2), and the analysis and synthesis criteria (Section 3.3).

3.1. Search strategy

Our methodology centers on the systematic identification, location, and evaluation of relevant literature using search strings and established search standards within a citation database. The process begins with defining the focus of the SLR, which serves as a preprocessing step. Based on this focus, we develop a structured keyword framework, which involves selecting appropriate keywords.

RQ1 aims to identify the main areas of airline or airport ground operations that focus on process optimization. We include both airline

and airport keywords because tasks may be attributed to either airlines or airports. Therefore, we initiate our SLR with a bibliometric study to identify the relevant keywords for airline or airport ground operations. Furthermore, since we mainly focus on the application of operations research techniques in the studies, we include keywords related to optimization, decision-making, forecasting, and so forth.

We follow the following steps for the bibliometric study.

1. We search for review articles related to the airline or airport ground operations discipline. We consider both the author and index keywords of all selected review articles and then export the RIS file.
2. Then we use a bibliometric network from VOSviewer, a bibliometric and network analysis tool to visualize the co-occurrences of keywords including the top 25 most frequently used keywords.
3. Next, we refer to the most prominent keyword sets and identify relevant clusters based on the strength of the association between them.
4. Finally, we remove the keywords related to document types and scientific characteristics from the bibliometric network.

We visualize the most common keywords through “VOSviewer”, see Fig. 3.

The network visualization reveals four interconnected research clusters. The clusters can be broadly segregated by focus: real-time decision-making under uncertainty (red), aircraft-level modeling and passenger behavior (green), optimization methods applied to concrete airport problems (blue), and system-wide integration and efficiency (yellow). The clusters are highly interconnected, suggesting that while different themes exist, the research domains overlap significantly. Furthermore, results indicate that various keywords are employed to describe ground operations, often using differing terminology. For example, some articles have used ground handling instead of ground operations. Therefore we accounted for all these keywords in our structure.

Below, we list the three groups that together define the keyword structure (Table 1). The first two groups mainly consider the application areas while the third group considers the optimization aspect. Finally, all three keyword groups are combined into one search string.

We choose Scopus, the largest abstract and citation database for peer-reviewed academic literature, as our source of study (see <https://www.elsevier.com/products/scopus>).

3.2. Selection criteria

The keyword structure that we introduced in Section 3.1 still requires pruning the pool of papers as not every article will contribute to

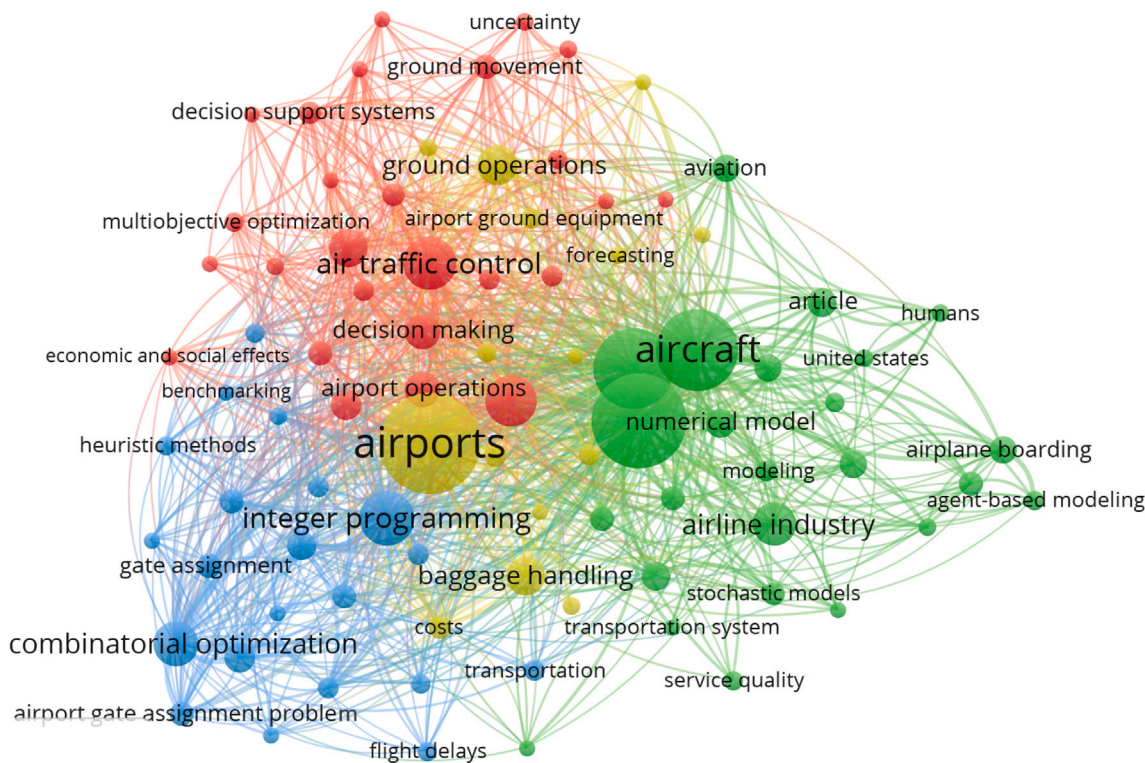


Fig. 3. Network visualization from VOSviewer.

Table 1
Keywords categorized into three groups.

Group 1	"airline" OR "airport"
Group 2	"ground operat*" OR "ground equipment" OR "ground movement" OR "ground handling" OR "ground support" OR "turnaround" OR "gate assignment" OR "boarding" OR "baggage handl*"
Group 3	"decision making" OR "optim*" OR "forecast*" OR "predict*" OR "schedul*" OR "decision support" OR "model*"

answering the research questions. According to Denyer and Tranfield (2009), the following four steps of inclusion/exclusion criteria can be used to derive an applicable paper pool. We appropriately modified each step to suit the specific needs of our study.

Step 1: Resulting articles should fulfill three inclusion criteria related to the document type itself:

1. The articles should be fully published in English
2. The articles' subject areas have to align with the academic fields taken into consideration; computer science, social sciences, mathematics, business management and accounting, decision sciences, economics, econometrics and finance, multidisciplinary
3. Only academic articles are included

We additionally define a time frame, restricting the selection to papers published between January 2004 and October 2024.

Step 2: The remaining article pool's abstracts are screened to identify the relevancy of the content. The following inclusion criteria are applied for further screening:

1. The methodologies used should be within the operations research domain.
2. Quantitative analysis needs to be carried out

Step 3: The following exclusion criteria were decided upon.

1. Remove all the duplicate articles keeping the most recent version

2. Remove the articles with the following specific keywords: "ground-based augmentation systems", "ionospher*", "air traffic control", "satellite*", "air navigation", "diseases", "viruses"

We identified several areas of literature that fall outside the scope of our study, including research on aircraft and air traffic navigation services, airport runway construction, and the impact of disease contamination beyond COVID-19. To maintain focus on our core research domain, we excluded these keywords from our analysis.

Step 4: After applying the inclusion and exclusion criteria, we download all the remaining articles for full-text reading. However, some articles may not be publicly accessible, or their content may lack sufficient quality for further analysis. Therefore, the final screening layer applies additional criteria to enhance the likelihood of successfully identifying and selecting relevant articles:

1. The articles should be available to access online
2. The article should be published in a peer-reviewed journal.

3.3. Analysis and synthesis criteria

After executing the above search strategy and selection criteria, we obtain a pool of papers that could answer the research questions. This collection of papers has been selected for data analysis followed by a bibliometric analysis. We use VOSviewer to analyze publications, authors, journals, and publishers. We conduct a further analysis of the main areas of airline ground operations that have received attention in the academic literature, along with their corresponding optimization methods. Finally, we discuss the future research directions corresponding to each main area. Fig. 4 provides a flow chart of the search strategy corresponding to the SLR methodology resulting in 199 papers.

4. Results

This section provides a comprehensive overview of the results of the research article pool retrieved from the search strategy presented

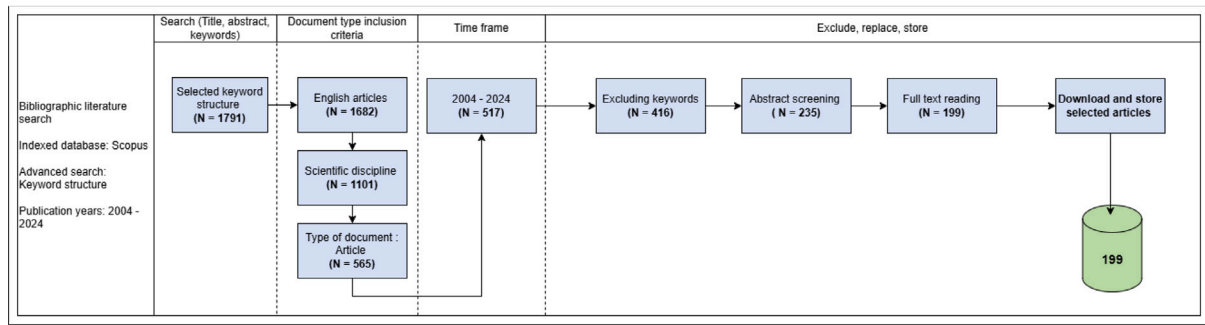


Fig. 4. Roadmap corresponding to the SLR methodology from Section 3.

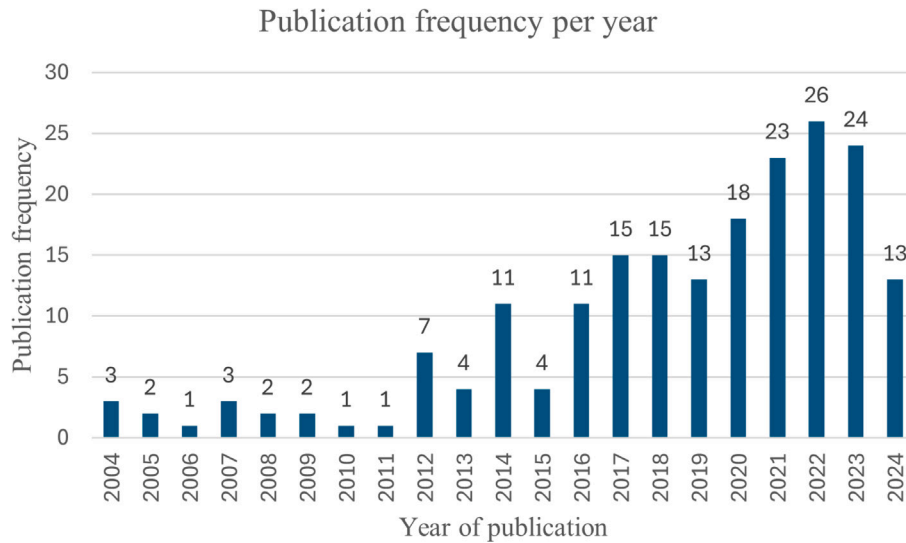


Fig. 5. Publication frequency of the 199 papers over the years.

in Section 3. First, we summarize key findings from the bibliometric analysis. Next, we provide an overview of the analysis where we define the categorization of the airport ground operations to answer RQ1.

Over the past decades, the number of applications in optimizing airport ground operations has gradually increased. Fig. 5 illustrates the publication frequency per year, showing a very small number of relevant yearly publications until 2012, after which the field experienced significant growth. From 2019, the figure shows a growth of publications included in the scope.

We selected the 20 journals with the highest frequency of publications from the pool of 199 selected papers as shown in Fig. 6. The majority of studies have been published in the Journal of Air Transport Management. The journals Transportation Research Part C: Emerging Technologies and European Journal of Operational Research exhibit the second and third-highest publication percentages.

The first research question is to identify the key functional areas. Therefore, we analyzed all publications and segregated them into relevant functional areas within airport ground operations. We identified seven key functional areas as shown in Fig. 7: airport gate assignment, boarding process and strategies, turnaround process, aircraft taxiing and ground movements, ground support equipment handling, baggage handling, and ground workforce planning and scheduling. Airport gate assignment dominates with 34% of the publications, reflecting its crucial role in optimizing airport throughput and minimizing delays, far surpassing other areas. In comparison, boarding processes and strategies follows at 17%, signifying its importance in minimizing aircraft turnaround time and improving overall operational efficiency. Although baggage handling (6%) and ground workforce planning and

scheduling (4%) are vital to operational success, the numbers are relatively small compared to other cases.

The bar chart in Fig. 8 provides an overview of the percentage of publications focused on key functional areas in airport ground operations per year. Airport gate assignment dominated research from 2004 to 2018, reflecting its critical role in operational efficiency, though its share slightly decreased in recent years. Boarding processes and strategies maintain strong attention, particularly from 2006 to 2009. However, interest gradually declined in the following years before experiencing a notable resurgence in 2019. Turnaround processes fluctuate over the years but gained significant attention post-2020, indicating a renewed interest in optimizing aircraft turnaround during recent years. Aircraft taxiing and ground movements and ground support equipment handling maintain a moderate but fluctuating focus. In contrast, baggage handling and workforce planning and scheduling remain the least explored areas, suggesting opportunities for future exploration. Although workforce planning and scheduling have been widely studied across industries (De Bruecker et al., 2015; Castillo-Salazar et al., 2016; Van den Bergh et al., 2013), literature specifically focused on airline ground operations remains limited. Unlike airport gate assignment, which has dedicated methodologies and substantial literature, workforce planning is often addressed as a general problem with applications in the airline domain. As a result, such contributions may not be prominently reflected in this SLR.

5. Discussion

This section analyzes the results related to RQ2 and RQ3. The key functional areas identified in Section 4 will be discussed in the

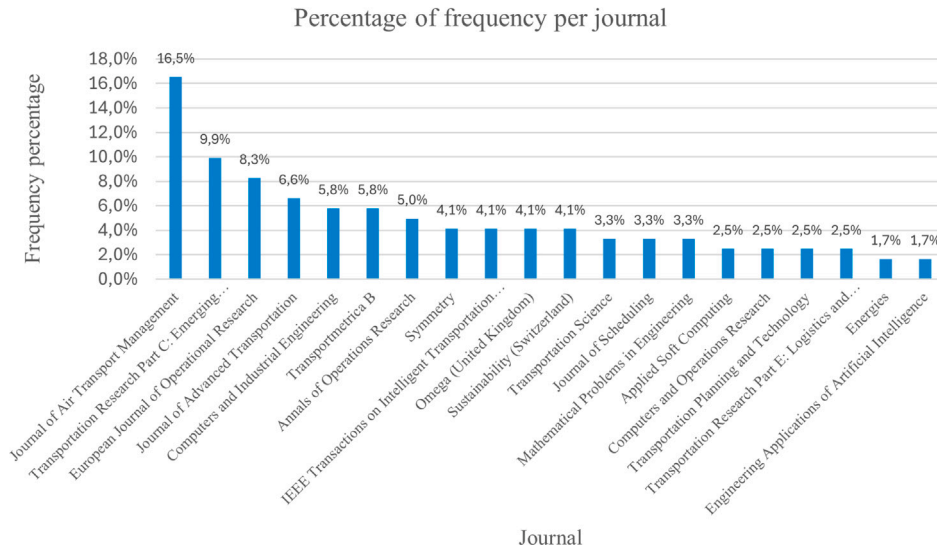


Fig. 6. Frequency percentage of highest published 20 journals.

Research distribution percentages of key functional areas in airport ground operations

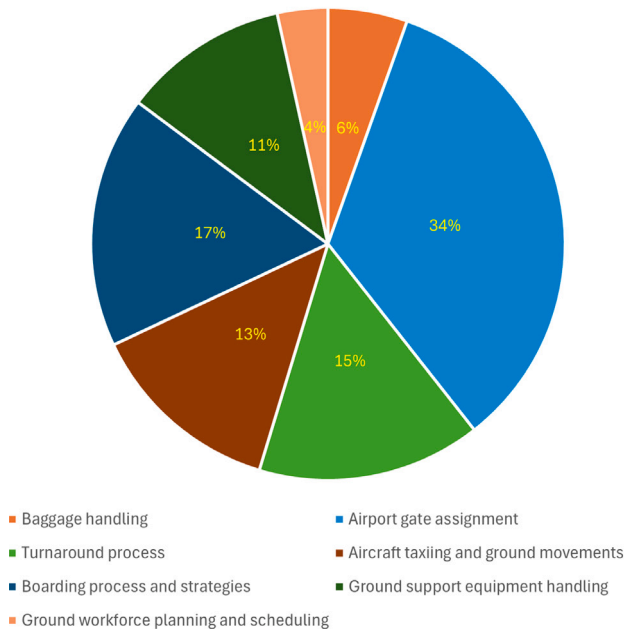


Fig. 7. Percentage distribution of papers across key functional areas in airport ground operations.

following subsections. Each area will be examined in terms of the problems studied and methodologies employed. Finally, we summarize the future research directions per area and highlight the overarching future research directions of the entire airport ground operations domain.

5.1. Airport gate assignment

The airport gate assignment problem (AGAP) is an essential part of airport ground operations, and has been extensively studied over the past two decades. The main aim of the airport gate assignment is to assign suitable gates for arriving and departing flights while maintaining schedule adherence as closely as possible. The AGAP is influenced by various factors, including the type of route, route distance, aircraft

type, aircraft turnaround time, and the walking distance for passengers between transferring flights (Wang et al., 2014). Multiple formulations of the AGAP have been proposed in the literature. We first discuss its basic modeling approach and the most commonly employed solutions. Subsequently, we discuss variations and extensions and modifications of the problem and explain how their modeling and solution strategies diverge from standard solutions.

One of the fundamental decisions in the AGAP is to minimize the number of aircraft assigned to gates or the apron at any given time while ensuring high efficiency and a reasonable arrangement. According to Karsu and Solyali (2023), the combination of a mixed-integer linear programming model and a metaheuristic solution method is an effective and easy-to-implement approach for solving such problems. Furthermore, to achieve a higher performance and avoid premature convergence, some literature suggests the use of evolutionary methods, such as the Wavelet Mutation Strategy Differential Evolution (Deng et al., 2021).

Crowded aprons and closely parked aircrafts can lead to taxiing conflicts and safety incidents, such as wing-tip scrapes between aircraft due to insufficient spacing. The basic problem has been extended to consider gate operational safety or apron operational safety as a constraint, to minimize the variability in gate idle time periods (Jiang et al., 2023c). The AGAP has also been explored with various additional considerations, such as gate closures and potential blocking of neighboring gates. For instance, Dorndorf et al. (2012) examine such restrictions, with the objectives of maximizing assignment preference, minimizing unassigned flights during overload, reducing tow operations, and maximizing a robustness measure while minimally deviating from a reference schedule. The problem is formulated as a clique partitioning problem and solved using an ejection chain algorithm, which is a heuristic optimization technique.

The bi-objective gate assignment problem represents another subclass of the basic AGAP. The commonly studied bi-objective optimization formulation seeks to simultaneously minimize the total walking distance of passengers and the total cost associated with the gate assignment. These problems, like the basic gate assignment problems, are frequently addressed using heuristic or metaheuristic techniques, but methods employing flow-based and column-based models like column generation can also produce optimal or near-optimal solutions (Daş and Gzara, 2024; Cai et al., 2019). Researchers have also investigated optimizing the AGAP to address both passenger experience and economic viability. By employing MILP models, Silva et al. (2023) have integrated passenger behavior modeling within the AGAP.

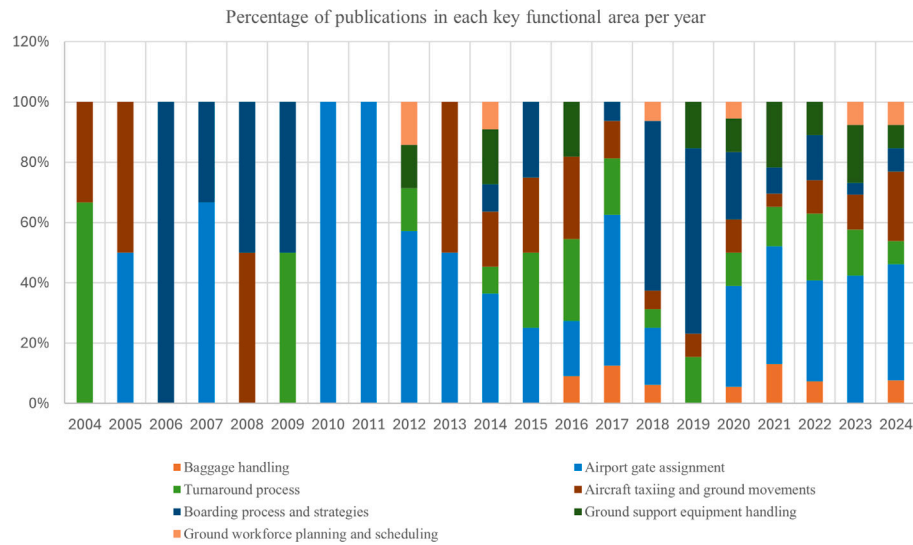


Fig. 8. Percentage of publications under each key functional area over 20 years.

The literature on AGAP has expanded from bi-objective optimization to include multi-objective optimization considerations. Common objectives in multi-objective optimization of gate assignments include minimizing passenger walking distance, maximizing the boarding bridge utilization rate, minimizing the number of towing activities, and minimizing fuel consumption of aircraft taxiing (Deng et al., 2024; Jiang et al., 2013; Li et al., 2024; Deng et al., 2017). To address these complex, multi-faceted problems, researchers have employed column generation-based algorithms and metaheuristic methods such as large neighborhood search and pareto simulated annealing. For large-scale problems, heuristic approaches are often recommended to improve solution speed (Bi et al., 2020). Additionally, hybrid methods, such as adaptive co-evolutionary ant colony optimization algorithms and quantum-inspired evolutionary algorithms based on decomposition mechanisms, have been proposed to further enhance the solution approach (Deng et al., 2024). Yuan et al. (2020) consider the gate assignment problem with three objectives: minimizing the number of ungated flights, reducing the total walking distance for transferring passengers, and minimize the variance in gate idle time. They employed a solution approach based on Lagrangian relaxation to effectively tackle this complex optimization problem. Additionally, the AGAP can be formulated to consider the objectives of assigning passengers to gates near shopping facilities and minimizing the total passenger walking distance (Daş, 2017). These multi-objective approaches aim to enhance the shopping revenues for airport operators.

The dynamic real-time AGAP has emerged as another extension of the basic problem. To address the dynamic nature and complexities of gate assignment in practice, researchers have proposed various modeling approaches. These include Markov decision processes (MDP) and deep reinforcement learning-based techniques (Jiaming et al., 2020), as well as two-stage hybrid algorithms combining imitation learning and genetic algorithms, which can significantly reduce the number of iterations and improve population convergence speed (Ding et al., 2023). Cheng et al. (2017) show that combining tabu search with path relinking strategies into a hybrid solution procedure leads to superior performance compared to other metaheuristics. Furthermore, a game theory-based approach has been developed by Zeunert and Herrich (2022). Additionally, Oussama and Abdellatif (2018) propose a time-dependent, multi-agent MDP model to support gate controllers in managing uncertainty. The model aims to reduce the need for online schedule adjustments by providing more robust decision-making capabilities.

Another key decision is the optimal reassignment of aircraft to gates in response to deviations in flight arrival and departure times, resolving conflicting gate assignments. Maharjan and Matis (2011) have utilized binary integer programming for optimal reassignment, demonstrating the speed and efficiency of this model in tackling real-world and real-time settings. In the context of optimal reassignment, neighborhood moves such as insertion, interval exchange, and time shifts are then incorporated into tabu search and memetic algorithms (Lim et al., 2005). Furthermore, branch-and-price algorithms, along with column generation accelerated by gate clustering, have been employed to solve these reassignment problems (Kim et al., 2023). These heuristic approaches have proven effective, exhibiting strong performance in small-scale instances and significantly outperforming the exact solvers in large-scale scenarios.

Robust gate assignment strategies that account for stochastic delays can mitigate the propagation of delays between flights via gates. Employing robust optimization methods is well-suited when the assignment is designed to constrain the probability of multiple flights being assigned to the same gate (L'Ortye et al., 2021; Diepen et al., 2012). Another important aspect is the relationship between schedule robustness and aircraft delay heterogeneity, which can be optimized through the strategic allocation of buffer times within gate schedules. To this end, Du et al. (2023) introduce two new robustness measures to quantify potential gate conflicts and model the Robust Gate Assignment Problem as a set-covering problem, minimizing both expected conflict costs and basic operating costs. A column generation-based heuristic is developed to solve large instances in reasonable time. Furthermore, Liu and Xiang (2023) have developed a model with a branch-and-price framework and a pulse algorithm to investigate the trade-offs between robustness and operational efficiency in the Robust Gate Assignment Problem.

Apart from dynamically assigning aircraft to gates, gates and their boarding bridges can also be dynamically reconfigured in order to be compatible with various aircraft types. This problem, driven by the continuous evolution of the fleet mix at airports and the increasing frequency of aircraft movements, has been another focus of study. This problem has been formulated as a mixed-integer model, and the solution is effectively addressed through a column generation-based heuristic approach (Hagspihl et al., 2022).

The AGAP has evolved to integrate both airside and landside constraints, such as the capabilities of facilities like check-in, security, and passenger transfers. New joint models have been developed that optimize the use of various airport resources. The AGAP now integrates the movement of passengers and freight within the terminal with the

taxiing of aircraft. The joint scheduling of airport gates and runways is one such integrated model, which aims to assess service quality, operational efficiency, and environmental impacts. Commonly, genetic algorithms have been utilized to solve these integrated problems (Hu et al., 2024). Another line of research focuses on jointly optimizing gate allocation and aircraft turnaround operations. For this, a hybrid metaheuristic algorithm based on the Shuffled Frog-Leaping Algorithm and the Grasshopper Optimization Algorithm has been used, outperforming standard mixed integer programming (MIP) solvers (Asadi et al., 2021). Additionally, a joint assignment problem for runways and gates, considering parking time differences and airline preferences, has been solved using a branch-and-price algorithm and a pulse algorithm. These joint assignment models can effectively reduce ground taxiing loss, optimize gate assignment robustness, and increase gate utilization (Jiang et al., 2024). For a focused literature review and detailed discussion of airport gate assignment problems, we refer the reader to Daş et al. (2020). Table 2 summarizes the literature on the AGAP, categorizing the related problems and the corresponding optimization techniques. The solution methods are classified based on their fundamental characteristics. Specifically, heuristic methods are distinguished from meta-heuristics, which include approaches such as evolutionary algorithms and fuzzy bee colony optimization. Additionally, learning-based approaches encompass techniques from machine learning and reinforcement learning. This classification is consistently applied across all summary tables of optimization techniques within each key functional area.

Despite methodological diversity in AGAPs, most models rely on simplifying assumptions that limit applicability in real-world airport environments. Several works incorporate stochasticity only partially, for example, by modeling flight delays or stochastic service times while still enforcing deterministic buffer times or fixed feasibility rules (Yan and Tang, 2007; Dorndorf et al., 2012, 2017). Others adopt static or discretized time windows, assuming known arrival and departure times and fixed service durations (Lim et al., 2005; Genç et al., 2012; Du et al., 2014), which prevents the schedule from adapting to real-time disruptions. Many formulations also treat gates as interchangeable resources, omitting key operational constraints such as gate reconfiguration, aircraft-type compatibility, or dynamic capacity adjustments (Maharjan and Matis, 2012; Karsu et al., 2021; Deng et al., 2021). Robustness is often addressed through deterministic factors such as fixed idle-time buffers or variance minimizing objectives rather than through explicit stochastic delay propagation (Dorndorf et al., 2012; Deng et al., 2017; Guépet et al., 2015). Moreover, most models assign each flight to exactly one gate in a static optimization setting without considering dynamic reassignment, recovery strategies, or real-time gate swapping under disruptions (Yan and Tang, 2007; Dell'Orco et al., 2017; Deng et al., 2021). As a result, assumptions such as independent flight tasks, deterministic service durations, and isolated gate operations limit the ability of existing AGAP models to capture cascading effects, equipment interactions, or operational uncertainties.

5.2. Boarding process and strategies

The aircraft boarding process, which starts from the first passenger's entry into the aircraft and ends with the final passenger's seating, represents a substantial component of the aircraft turnaround time and significantly impacts the overall operational expenses incurred by airlines. Academic literature has proposed a variety of boarding methods aimed at minimizing the time required to board aircraft. As the total passenger boarding time frequently exceeds the duration of refueling and restocking, optimizing this process could lead to substantial cost savings for individual airlines, particularly for aircraft making multiple flights per day (Steffen, 2008). There are several boarding strategies used in boarding models. We first explain the common boarding strategies in the literature, then the extended boarding strategies, and finally the factors that influence the efficiency of the boarding process.

Several boarding strategies are discussed in the literature focused on improving the efficiency of the boarding process. One of the methods is the free boarding, which means that passengers are not given assigned seats in advance and they choose any available seat on a first-come, first-served basis. Qiang et al. (2018) employ a simulation framework based on a cellular automaton model to model passenger flow dynamics in the free boarding process and a logit-based discrete choice model is used to capture passenger seat selection behavior. The influence of sensitivity parameters on the mean boarding time is analyzed to understand different behavioral patterns and the new model can qualitatively replicate key dynamics of the free boarding process. Random boarding is also explored in the literature, where passengers board in no particular order, often used by airlines that do not assign seats but divide passengers in groups based on check-in order or priority status. Delcea et al. (2018c), and Bachmat et al. (2009) study the random seat boarding method compared with common boarding practices and used agent-based simulation and discrete-event simulation respectively to assess the boarding times. In addition, priority boarding is a widely implemented strategy, though it can introduce inefficiencies when priority passengers board randomly and disrupt the intended sequence, causing aisle blockages. Therefore, Kisiel (2020) analyzes the sensitivity of most common boarding strategies to the number of priority passengers and uses a discrete-event simulation model to choose a suitable strategy.

The reverse pyramid method is another approach discussed in the literature, which corresponds with boarding passengers in groups from the back of the plane to the front while prioritizing window seats first, and boarding aisle seats last. Cotfas et al. (2019) and Delcea et al. (2018c) use the reverse pyramid boarding method, aiming to balance boarding time and efficiency. Agent-based simulation is used to evaluate the solutions and the best solution is determined through local search optimization. Another strategy is the outside-in boarding method in which passengers with window seats board first, followed by those with middle seats, and finally those with aisle seats. Luo et al. (2021) propose two strategies to improve the boarding process. The first was an enhanced outside-in approach that allows two aisles to board at the same time, though it requires passengers to queue in advance. The second was a deterministic queue-ordered method, a boarding strategy where passengers are assigned a specific order in which they are expected to board the aircraft.

Literature also suggests novel boarding strategies such as dynamically optimized boarding, a flight-specific strategy that minimizes boarding interference by organizing passengers into cliques based on seating preferences to reduce the aircraft boarding time. Zeineddine (2017) apply this method and demonstrated that it effectively addresses passengers' crucial need to board in cliques. Furthermore, it offers a marginally enhanced performance relative to the Window-Middle-Aisle (WilMA) strategy, and a significant improvement over commonly employed approaches such as random boarding. Another extension is individual-based boarding approach. Qiang et al. (2014) introduce an improved aircraft boarding strategy that assigns passengers to seats based on the amount of luggage they carry, with passengers with the most luggage boarding first. To evaluate the performance, a sophisticated simulation environment using a cellular automaton model was developed. Further, Schultz and Schmidt (2018) compare two individual-based approaches, denoted by paxSim & PAXelerate, which are simulation frameworks that cover both specific passenger behavior during boarding and operational airline constraints. Besides the common methods used, there are extended methods, like "Pedestrian Following model" from Tang et al. (2018), which incorporates group behavior and the quantity of luggage into the boarding model. The quantity of luggage may make each passenger's boarding behavior more complex, while group behavior can shorten some passengers' delay time and the total boarding time.

The COVID-19 pandemic has resulted in another perspective on boarding and seating passengers. Salari et al. (2022) study assigning passengers to seats while considering two social distancing measures

Table 2
Summary of optimization techniques in AGAP and related problems.

Problem/Focus Area	Solution Method	Paper
AGAP on operational efficiency and feasible spatial arrangement	Heuristics	Genç et al. (2012), Yin et al. (2013), Guépet et al. (2015), Wang et al. (2014), Karsu et al. (2021), Liang et al. (2020), Xiao et al. (2020), Deng et al. (2021), Y. Li et al. (2021), H. Li et al. (2021), Stollenwerk et al. (2020), Karsu and Solyalı (2023)
	Meta-heuristics	Dell'Orco et al. (2017), Chow et al. (2022)
	Learning based approach	Li et al. (2022)
AGAP with various additional constraints	Exact	Bi et al. (2022), Jiang et al. (2023c)
	Heuristics	Dorndorf et al. (2012), Ornek et al. (2022)
Bi-objective AGAP	Exact	Maharjan and Matis (2012), Daş and Gzara (2024)
	Heuristic	Silva et al. (2023)
	Meta-heuristics	Cai et al. (2019)
Multi-objective AGAP	Exact	Jiang et al. (2013)
	Heuristic	Bi et al. (2020), Li et al. (2024)
	Meta-heuristics	Nikulin and Drexel (2010), Deng et al. (2017, 2018), Daş (2017), Kaliszewski et al. (2017), Zhang et al. (2022), Jiang et al. (2023a), Deng et al. (2024)
Dynamic/ real-time AGAP	Exact	Zeunert and Herrich (2022)
	Learning based approach	Oussama and Abdellatif (2018), Jiaming et al. (2020)
	Heuristics	Yan and Tang (2007), Dorndorf et al. (2017), Hagspihl et al. (2022)
	Meta-heuristics	Cheng et al. (2017), Hu et al. (2021), Ding et al. (2023), Xiao et al. (2023)
Optimal reassignment of aircraft to gates	Exact	Maharjan and Matis (2011)
	Heuristics	Yuan et al. (2020), Kim et al. (2023)
	Meta-heuristics	Lim et al. (2005)
Robust AGAP	Exact	L'Ortye et al. (2021)
	Heuristics	Dorndorf et al. (2007), Diepen et al. (2012), Kim and Feron (2017), Du et al. (2023), Liu and Xiang (2023)
	Meta-heuristics	Kim and Feron (2013), Zhao and Cheng (2014), Liu et al. (2016), Zhao et al. (2018), Tan and He (2021), She et al. (2022)
Joint optimization of the AGAP with other optimization problems	Heuristics	Jiang et al. (2024), Yang and Gao (2022), Liu et al. (2023), Du et al. (2014)
	Meta-heuristics	Hu et al. (2024), Behrends and Usher (2017, 2016), Jiang et al. (2023b), Asadi et al. (2021)

by using a MIP model. Their results show that more passengers can be assigned safely to seats when family groups are seated near others within the same group. Additionally, Qureshi and Qureshi (2022) use discrete-event simulation to evaluate the impact of COVID-19 policies on the passenger boarding processes. Steffen's method, which involves boarding window seat passengers starting from the back row and then proceeding with window seat passengers two rows ahead of the last boarded passenger, is highly effective at reducing COVID-19 risk but challenging to implement. In contrast, the reverse pyramid and WilMA are easier to implement although they are less effective.

Improving the prediction of boarding time and monitoring boarding progress has become another focus in recent research. One such approach is the time-aware boarding strategy proposed by Willamowski and Tillmann (2022). They develop a MIP, which is solved using a local search heuristic method, and they show that for several airplane cabin layouts, high-quality or even optimal solutions can be obtained within reasonable time. Kuo (2015) proposes a new integer programming method to minimize the time required to eliminate the interferences among all passengers when boarding a flight. Wittmann (2019) and Fabrin et al. (2024) present agent-based approaches to

reduce the boarding time while taking into account passenger satisfaction. Some studies provide an approach for real-time evaluation of the boarding process using the capabilities of a sensor equipped cabin. Schultz (2018b) describes a set of indicators and a method for real-time assessment of the aircraft boarding progress, which is a crucial step in accurately forecasting aircraft boarding duration. It utilizes the aircraft seats as a sensor network to detect seat occupancy status, which serves as the basis for calculating an interference potential metric to estimate boarding time.

The boarding strategies discussed above primarily utilize jet bridges. In addition, the literature discusses the use of apron buses with relevant boarding strategies. Milne et al. (2020) study boarding of a two-door airplane using two apron buses and proposed a MIP-based method that reduces the boarding time when compared with the baseline approach that assigns groups of passengers with seats furthest from one of the airplane doors to the first apron bus and assigns remaining groups to the second apron bus. Delcea et al. (2019) propose 15 new methods for the boarding of a two-door airplane using two apron buses that are tested against the back-to-front method. Experimental results show that the most effective proposed methods combine elements of the WilMA and the Reverse Pyramid method, specifically adapted for apron bus

boarding. A variation on this topic arises when the aircraft is only partially occupied. Milne et al. (2019a) solve this problem using a greedy method, which reduced boarding time almost by half when compared to the boarding methods commonly used in many airports.

We further explore the factors that influence the efficiency of the boarding process. One commonly identified factor is the impact of hand luggage and passenger behavior. Schultz and Schmidt (2018) indicate a decreasing boarding time when the number of hand luggage items is reduced and when passengers with large bags sit in the rear cabin. Furthermore, some studies explain that the quantity of luggage can complicate each passenger's boarding behavior, while group behavior may help reduce some passengers' delay times and shorten the overall boarding time (Tang et al., 2018). Finally, some models improve the utilization of the available storage space for hand luggage and increase satisfaction levels due to the faster luggage storage process during boarding (Ren et al., 2020). Typically, Markov chain Monte Carlo optimization algorithms, agent-based simulation models, and computer simulation have been used to solve these problems. For a comprehensive review of airplane boarding methods, we refer the reader to a survey by Jaehn and Neumann (2015) for a classification scheme, Bidanda et al. (2017) for a review on optimization models, and Kobbaey and Bilquise (2022) for an analysis using agent-based simulation. Table 3 summarizes the literature on the boarding process and strategies, categorizing the related problems and the corresponding optimization techniques. The first column presents the boarding strategies and the related problems, while the second column outlines the evaluation methods used for the strategies and the solution methods applied to address the associated problems.

The analysis of literature on the boarding process and strategies shows that, despite methodological variety, the contributions share several simplifying modeling assumptions that constrain real-world applicability. Many models treat boarding as a one-dimensional, forward-moving discrete process with homogeneous dynamics (Schultz, 2018a), assume instantaneous or perfectly reliable seat-status information (Schultz, 2018b), or rely on generic luggage-handling distributions and fixed aircraft layouts (Qiang et al., 2014; Tang et al., 2018). Service-time elements such as luggage stowing, seat-position and interpersonal interference are often deterministic or governed by simple distributions (Steffen, 2008; Bazargan, 2007), while agents typically follow fixed rational rules without modeling human variability, group behavior, counter-flows, or non-compliance (Delcea et al., 2018b). One study further imposes static, pre-defined boarding orders generated at check-in, assuming no late arrivals, equipment failures, or gate re-sequencing (Zeineddine, 2017). Additionally, asymptotic formulations omit aircraft-specific features, seat-preference behaviors, and stochastic disruptions (Bachmat et al., 2009, 2006). Collectively, these assumptions lead to limitations such as the inability to represent aircraft-type heterogeneity, multi-door configurations, seat-position interferences, or real-time operational uncertainties that are critical aspects of modern boarding operations that merit explicit modeling and discussion.

5.3. Turnaround process

Aircraft turnaround consists of all necessary activities for preparing an aircraft for the next departure. Aircraft turnaround activities need to be processed with strict requirements, consisting of interdependent activities and various personnel. The ground operations process can be divided into three categories based on distinct service objectives, namely services for passengers, aircraft, baggage and cargo. Each operation has its own time and sequencing constraints. Turnaround time is defined as the time an aircraft is on the ground between two consecutive flight legs. The entire turnaround time can change depending on the aircraft type, the number of passengers, the quantity of cargo, the efficiency of other turnaround activities, and operational constraints.

The efficiency of aircraft turnaround operations and turnaround time, as well as their performance under various conditions, has been explored in the literature. The critical path method (CPM), which breaks down the process into smaller tasks and identifies dependencies, has been applied in various studies. Tang et al. (2024) use this method within a discrete-event simulation model to assess the impact of COVID-19 on ground operations, comparing the turnaround process before and after the pandemic. The analysis highlights the most affected subprocesses, guiding airports on areas to prioritize. However, CPM fails to account for uncertainties, which is addressed by Wu and Caves (2004a,b) through the use of monte carlo simulation and discrete-event simulation techniques. Their study identifies two key factors affecting flight punctuality: buffer time in ground operations and operational efficiency. A follow-up study by the same authors demonstrates that optimizing buffer time enhances punctuality while reducing system costs (Wu and Caves, 2004b). Similarly, Mota et al. (2017) apply discrete-event simulation to model turnaround operations at a European airport, finding that higher staffing levels result in faster turnaround times. Abd Allah Makhloof et al. (2014) develop a rule-based system to manage flight preparation during turnaround, using Project Evaluation and Review Technique to improve operational efficiency and ground activity allocation. Furthermore, Rott et al. (2023) apply process mining techniques within the airport collaborative decision-making framework, which enhances efficiency through improved stakeholder coordination, to increase operational transparency and demonstrate its potential in optimizing airport operations. The implementation of new technologies has opened avenues for further improving the efficiency of turnaround operations. Conde et al. (2022) demonstrate how deploying the Airport Digital Twin at Aberdeen International Airport enhances the efficiency of flight turnarounds, reducing delays. Similarly, Yıldız et al. (2022) investigate deep learning-based systems for monitoring and managing airport ground services, significantly reducing delay-associated costs and improving overall operational performance.

Efficient operational procedures rely heavily on the optimal utilization of resources. Ground handling vehicle scheduling and routing problems have been a key focus in this context, with solutions often derived from heuristic methods (Padrón and Guimarães, 2019). For example, Guimarães and Padrón (2022) propose a stochastic approach for modeling the routing problems encountered during the scheduling of each ground operation activity and embedded monte carlo simulation within the CPM approach. Du et al. (2016) address the strategic fleet composition problem for towing tractors at airports. They present a set-covering formulation to optimize fleet size and mix, and a column generation heuristic is used to solve the model.

Turnaround operations are also significantly affected by unexpected events, which can disrupt airside operations and increase turnaround times. To address these challenges, researchers have developed dynamic critical path analyses and discrete-event simulation models, and tested those at medium and large airports (Asadi and Fricke, 2022; Nosedal Sánchez and Piera Eroles, 2018; Postorino et al., 2020). Disruptive factors such as ground handler strikes can lead to cascading delays. Malandri et al. (2019) highlight how reductions in ground operations staff below critical thresholds can exceed buffer times, necessitating emergency actions and contingency planning. Additionally, Gladys et al. (2022) explore the environmental impact of such disruptions on turnaround operations. Robust planning of turnaround activities across multiple service providers has been explored to ensure coordination and efficiency. Gök et al. (2023) investigate central scheduling approaches that integrate simulation optimization and constraint-based feedback routines. For instance, Glomb et al. (2023) propose a mixed-integer programming approach combined with a decomposition algorithm to jointly optimize turnaround models and aircraft assignments, addressing the complexities of real-time disruptions.

Closely linked to the above challenges, aircraft delay propagation and its connection to turnaround operations have received significant

Table 3
Summary of boarding strategies optimization.

Boarding strategy/ Problem	Evaluation or Solution Methods	Paper
Free boarding	Agent-based simulation	Qiang et al. (2018)
Random boarding	Agent-based simulation	Delcea et al. (2018c)
	Discrete-event simulation	Bachmat et al. (2009)
Priority boarding	Discrete-event simulation	Kisiel (2020)
Reverse pyramid method	Agent-based simulation	Cotfas et al. (2021), Delcea et al. (2020)
Outside-in strategy	Discrete-event simulation	Luo et al. (2021)
Dynamically optimized boarding strategy	Agent-based simulation	Zeineddine (2017)
	Discrete-event simulation	Schultz (2018a)
Individual-based boarding approach	Agent-based simulation	Qiang et al. (2014), Schultz and Schmidt (2018)
Extended boarding models	Mathematical modeling and simulation	Bachmat et al. (2006), Bachmat (2019)
	Discrete-event simulation	Tang et al. (2018)
Boarding strategies with COVID-19 impact	Heuristics	Salari et al. (2022)
	Discrete-event simulation	Qureshi and Qureshi (2022)
Boarding progress and time prediction	Regression and statistical methods	Hutter et al. (2019), Neumann (2019)
	Agent-based simulation	Delcea et al. (2018b), Schultz (2018b), Wittmann (2019), Fabrin et al. (2024)
	Mathematical modeling and simulation	Bazargan (2007), Kuo (2015), Willamowski and Tillmann (2022), Bachmat et al. (2023)
Use of Apron buses in boarding process	Heuristics	Milne et al. (2019a, 2020)
	Agent-based simulation	Delcea et al. (2018a), Cotfas et al. (2019), Delcea et al. (2019), Milne et al. (2019b)
Impact of luggage on boarding	Monte carlo simulation	Steffen (2008)
	Agent-based simulation	Nugroho and Asrol (2022)
	Regression and statistical methods	Ren et al. (2020)

attention in recent academic literature. Evler et al. (2022) address how to holistically model network delay propagation for airline schedule recovery using a heterogeneous vehicle routing problem (VRP) with time windows with an extended version of the resource-constrained project schedule problem. This approach, which allocates scarce resources to turnarounds, demonstrates how incorporating recovery options can improve network resilience and reduce costs. Similarly, D'Ariano et al. (2012) explore rescheduling and rerouting decisions to minimize delay propagation using branch-and-bound algorithms, while Evler et al. (2021a) assess schedule recovery strategies under uncertain arrival times, employing heuristic methods such as genetic algorithms. For an introduction to aircraft ground operations, with a focus on aircraft turnaround and passenger processes, we refer to Schmidt (2017). Table 4 summarizes the literature on turnaround process, categorizing the related problems and the corresponding optimization techniques.

Although the reviewed studies provide valuable simulation and optimization approaches for turnaround processes, they rely on several simplifying modeling assumptions. Many works adopt deterministic service times, which suppress operational variability and underestimate delay propagation (Evler et al., 2021b; Chen et al., 2015; Kellenbrink and Helber, 2015). Even when stochastic durations are used, they often depend on pre-selected distributions with independence assumptions that ignore correlations between sub-processes (Mota et al., 2017; Malandri et al., 2019). Another common assumption is a fixed or static process structure, where turnaround tasks are represented through predefined activity networks with hard precedence relations and limited flexibility for conditional operations (Kuster et al., 2009; Abd

Allah Makhloof et al., 2014). Several studies additionally rely on static time windows and exogenous schedules, assuming full knowledge of arrival times or fixed ground-handling configurations, thereby omitting dynamic reallocation and real-time adaptation (Chen et al., 2016; Wu and Caves, 2004a). Resource availability is often simplified through aggregate capacity pools, which risks overlooking bottlenecks arising from GSE movement, stand access, or multi-flight coupling (Evler et al., 2021b; Wu and Caves, 2004b). Collectively, these assumptions, deterministic timing, independence of tasks, static windows, and limited resource realism, suggest that existing models capture an idealized turnaround environment. As real operations exhibit uncertainty and interdependencies, recognizing such limitations is vital when interpreting reported performance gains.

5.4. Aircraft taxiing and ground movements

Airport ground movement areas are categorized into two distinct zones: the maneuvering area and the movement area (Park and Kim, 2023). The maneuvering area encompasses the runway and taxiway infrastructure utilized for aircraft takeoff, landing, and ground mobility, excluding the apron. The movement area consists of the apron and aircraft parking stands. Within the airport movement zones, all aircrafts transit along designated taxiways and taxi lines to facilitate their movement, collectively referred to as the taxiway network. The taxiing process involves two main stages: departure and arrival. For departure, after preparation, the aircraft is pushed back from its parking spot and then taxis along the taxiways to the runway, where it receives takeoff

Table 4
Summary of optimization techniques in turnaround operations.

Problem/Focus area	Solution Methods	Paper
Performance and efficiency of aircraft turnaround	Simulation	Wu and Caves (2004a,b), Khumboon and Isaradech (2016), Mota et al. (2017), Scala et al. (2017), Conde et al. (2022), Tang et al. (2024)
	Regression and statistical methods	Malighetti et al. (2023), Rott et al. (2023)
	Exact	Abd Allah Makhloof et al. (2014)
	Heuristics	Chen et al. (2015)
	Learning based approach	Yıldız et al. (2022)
	Rule based system	Karapetyan et al. (2017)
Turnaround resource scheduling	Heuristics	Du et al. (2016), Padrón and Guimarães (2019), Guimarães and Padron (2022)
	Learning based approach	Mamdouh et al. (2020)
Impact of unexpected events on aircraft turnaround	Exact	Glomb et al. (2023)
	Heuristics	Asadi and Fricke (2022)
	Simulation based optimization	Gładys et al. (2022), Gök et al. (2023)
	Simulation	Nosedal Sánchez and Piera Eroles (2018), Malandri et al. (2019), Postorino et al. (2020)
Delay propagation & turnaround	Heuristics	Evler et al. (2022), D'Ariano et al. (2012), Kellenbrink and Helber (2015), Kuster et al. (2009), Evler et al. (2021b)
	Simulation	Evler et al. (2021a)
	Regression and statistical methods	Kim and Bae (2021)

clearance. For arrival, the aircraft lands on the runway, taxis along the main and parking-area taxiways, and reaches its designated parking spot.

Most studies in this area focus on the ground movement problem, which involves efficiently moving aircraft between locations within an airport, such as gates/stands and runways, while minimizing time and fuel consumption (Guépet et al., 2017; Weiszer et al., 2022; Beke et al., 2023; Ravizza et al., 2014a). Researchers use various optimization techniques, including genetic algorithms, memetic approaches, and graph-based algorithms, to improve efficiency by considering factors such as stand holding and multi-graph constraints (Ravizza et al., 2014a; Beke et al., 2023). Other studies focus on conflict-free operations and trajectory optimization, with methods like dynamic routing and timing algorithms used to optimize taxi trajectories and reduce delays (Dabachine et al., 2020; Zhang et al., 2016). Real-time data integration and dynamic modeling play an important role (Weiszer et al., 2015; Obajemu et al., 2021). These papers highlight the use of high-fidelity aircraft models and real-time data processing, combined with genetic algorithms and active routing approaches, to optimize trajectories and routing in dynamic environments, improving decision-making. Additionally, multi-objective optimization is explored in some studies to balance time, cost, and sustainability using the methods to solve the quickest path problem and multi-objective routing and scheduling algorithms (Chen et al., 2016; Weiszer et al., 2020).

Airport taxi planning problems involve optimizing the movement of aircraft on airport taxiways to support efficient arrival sequencing, departure push-back management, and gate assignment while minimizing taxi time and avoiding conflicts (Marin, 2013). There are several methods used to solve this problem, including exact methods such as Lagrangian decomposition (Marin, 2013), and branch-and-bound methods (Marín and Codina, 2008), as well as heuristic methods (Weiszer et al., 2015; Chen et al., 2016) and simulation based optimization (Gołda et al., 2019).

Aircraft taxi route assignment is another studied problem, which involves the process of assigning paths for arriving flights from the runway to the parking gate or vice versa for departing flights (Wang

et al., 2024). Wang et al. (2024) propose a novel approach that reduces the problem scale through a spatial-temporal graph neural network and model taxi route assignment using a MILP. Furthermore, studies investigate the optimal routing strategies to minimize waiting times of aircraft while specifically addressing the stochasticity of arrivals (Bäuerle et al., 2004).

Several studies combined gate assignment (covered in Section 5.1) and taxiway planning. Yang and Gao (2022) propose a new scheduling model for the ground movement problem integrated with the taxiway routes' assignment and the gate allocation concerning both uncertainties by arrival time and off-block time information. Furthermore, Liu et al. (2023) develop an integrated model in a discrete time-space network to simultaneously deal with gate assignment and taxiway planning. Most of these studies employ integer linear programming to model the problem, which is subsequently addressed using solution methods such as genetic algorithms, branch-and-bound algorithms or local search methods like variable neighborhood descent.

Sustainable taxiing has emerged as another key area of research. Chen et al. (2024) construct a bilevel spatial and temporal optimization model of aircraft taxiing aim at improving operational efficiency while reducing aviation emissions. To solve the model, they employ simulation experiments in conjunction with genetic algorithms. A related advancement in this domain is the electrification of taxiing operations, which seeks to minimize the environmental footprint of airports. In this context, Ahmadi and Akgunduz (2023) that assigns electric-powered tow-tractors to aircraft, enabling taxiing with zero jet-fuel consumption.

Forecasting of taxiing times is a further focus area in the literature. The accurate prediction of taxiing times of arrivals and departures helps with runway sequencing, gate assignment, and ground movement. Therefore, researchers have adopted several approaches to estimate taxi times such as multiple linear regression analysis (Ravizza et al., 2013), log-linear regression analysis (Lordan et al., 2016), and soft-computing based regression methods (Ravizza et al., 2014b). Furthermore, Brownlee et al. (2018) propose a fuzzy rule-based system to estimate taxi times and their uncertainties. We refer the reader to Atkin et al. (2010) for a detailed overview on the airport ground movement problem. Table 5 summarizes the literature on aircraft taxiing and ground

Table 5
Optimization techniques for aircraft taxiing and ground movements.

Problem/Focus area	Solution methods	Paper
Ground movement problem	Heuristics	Dabachine et al. (2020), García et al. (2005), Ravizza et al. (2014a), Zhang et al. (2016), Chen et al. (2016), Guépet et al. (2017), Weiszer et al. (2020), Obajemu et al. (2021), Weiszer et al. (2022), Beke et al. (2023)
	Simulation based optimization	Gołda et al. (2019), Weiszer et al. (2015)
Airport taxi planning	Exact	Marín and Codina (2008), Marín (2013)
	Heuristics	Dabachine et al. (2020), Weiszer et al. (2015), Chen et al. (2016), Zhang et al. (2016), Weiszer et al. (2020)
	Simulation based optimization	Gołda et al. (2019)
Aircraft taxi route assignment	Learning based approach	Wang et al. (2024)
	Exact	Bäuerle et al. (2004)
	Heuristics	Idrissi et al. (2022)
Combined taxiway routing and gate allocation	Exact	Yang and Gao (2022), Liu et al. (2023)
	Heuristics	Behrends and Usher (2017), Jiang et al. (2023b)
Sustainable taxiing	Heuristics	Chen et al. (2024)
	Exact	Ahmadi and Akgunduz (2023)
Forecasting taxi times	Regression and statistical methods	Ravizza et al. (2013, 2014b), Lordan et al. (2016)
	Rule based system	Brownlee et al. (2018)

movements, categorizing the related problems and the corresponding optimization techniques.

Despite the diversity of optimization strategies adopted in the literature, several common modeling simplifications are evident across the studies. First, taxi and service times are predominantly treated as deterministic inputs, often derived from constant-speed assumptions, perfect predictions, or fixed segment-level speed profiles (Marín and Codina, 2008; Ravizza et al., 2014a; Chen et al., 2016; Brownlee et al., 2018; Weiszer et al., 2020). While some papers employ more elaborate estimation methods, such as fuzzy travel-time prediction or detailed speed-phase decomposition, these still reduce to static parameters rather than capturing stochastic fluctuation in pushback delays, congestion, or pilot behavior (Ravizza et al., 2014b; Brownlee et al., 2018; Weiszer et al., 2020). Interactions between aircraft are typically reduced to deterministic capacity or separation constraints, treating tasks as largely independent. This simplification ignores realistic dependencies such as queuing, shockwaves, and delay propagation (Marín and Codina, 2008; Ravizza et al., 2014a; Weiszer et al., 2015; Guépet et al., 2017). Furthermore, most formulations rely on static time windows, typically under time-expanded networks or single planning horizons (García et al., 2005; Brownlee et al., 2018), which limits adaptability to dynamic operational conditions. While these assumptions aid tractability, they risk overstating performance in real settings marked by uncertainty and interdependence.

5.5. Ground support equipment handling

Ground operations handling vehicles and equipment, or ground support equipment (GSE), are used to provide services to aircraft upon their arrival at the airport, including the apron, terminal, and other areas, in preparation for the next flight. GSE are predominantly utilized on the airside to provide a variety of services to aircraft, such as refueling, loading luggage, transporting passengers, loading food and water, and de-icing planes. There are several types of GSE, such as

aircraft push-back tractors, baggage tractors, belt loaders, container loaders, fuel trucks, ferry vehicles, etc. This highlights the need for optimization approaches to improve the performance, allocation, and scheduling of GSEs.

The optimal scheduling of special purpose vehicles for ground support services has been examined in the literature. It is commonly modeled as an integer programming model and solved as a network flow problem (Zhao et al., 2021; Han et al., 2020). Zhao et al. (2019) explore the collaborative scheduling of airport ferry vehicles and tractors using a bi-objective MIP model. They solved it using a lexicographic method and the Pareto method, based on particle swarm optimization. Furthermore, studies have explored minimizing the travel time between nodes in the airport ferry network using the shortest path algorithm (Han et al., 2020). Another line of research focuses on the airport baggage transport vehicle scheduling problem which is influential to the quality of airport service. Guo et al. (2020) introduce a genetic algorithm for the scheduling of airport baggage transport vehicles.

The diverse properties of ground-handling vehicles create scheduling and coordination challenges. Additionally, the uncertainty inherent in the service process further complicates optimization. Zhu et al. (2022) integrate refuellers and shuttle buses to provide a more realistic and coordinated optimization schedule for the dispatching of ground-handling vehicles. Multi-objective MIP and chance constraints are used to describe the uncertainty of flight arrivals. Padrón et al. (2016) investigate the broader problem of scheduling diverse vehicles for various ground operation activities, modeling it as a VRP with time windows. They employ a bi-objective algorithm driven by a workcenter-based decomposition strategy to address this challenge. Liu et al. (2024) propose the application of a combined simulation and optimization approach to effectively address the scheduling challenge due to the uncertainty of the service process of multiple vehicles.

Building on the foundational VRP-based approaches, recent studies have proposed several extensions to better capture the operational complexities of ground-handling vehicle scheduling. Zhao et al. (2023)

formulate the skill capacitated VRP with time windows, utilizing a vehicle scheduling model aimed at minimizing vehicle use and balancing workloads. The solution combines a genetic algorithm with large neighborhood search to optimize scheduling. Furthermore, [Andreatta et al. \(2016\)](#) address the VRP with hard time windows and stochastic travel times, proposing a branch-and-price procedure to integrate expected penalty costs for disruptions. [Y. Wu et al. \(2023\)](#) model airport ground operations as a multiple-fleet VRP with constraints like precedence and time windows, employing reinforcement learning to construct high-quality routing solutions efficiently, demonstrating strong generalization to larger and real-time instances. Scheduling multiple ground operations for various flights, each handled by a specific vehicle fleet, is a complex challenge. [Zhou et al. \(2023\)](#) address this by representing the operations as a VRP and formulating it as a MILP model. They leveraged a learning-assisted large neighborhood search method with imitation learning and graph convolutional networks, which improved scalability and performance in real-world scenarios.

Recently, researchers have focused on integrating electric ground support equipment (e-GSE) into airport operations to reduce emissions and improve efficiency. Some studies investigate the scheduling problem for a mixed fleet of fuel and electric vehicles, developing a mixed operation model with time windows that minimize costs associated with time, energy, and emissions ([Bao et al., 2023](#)). The solution employs an adaptive large neighborhood search algorithm. Another study develops a conceptual model for implementing e-GSE, considering factors like emission standards, infrastructure requirements, and compliance with government regulations ([Sari et al., 2021](#)). Another aspect of the integration of e-GSE is forecasting energy demand in electrified ground operations. [Ajayi et al. \(2023\)](#) use a machine-learning-based model for energy demand forecasting and highlights the accuracy in predicting energy needs, enabling efficient energy management and planning for future electrified airport operations. Another study has looked into the cost-effectiveness of implementing e-GSE in comparison to diesel-powered GSE using the benefit-cost ratio method ([Saria et al., 2022](#)).

Another line of research has focused on integrating real-time tracking and efficient scheduling of GSE, beyond just support vehicles, to enhance ramp safety and operational efficiency. [S. Wang et al. \(2020\)](#) address the challenge of optimizing the scheduling of GSE and detecting ramp risks by utilizing real-time kinematic tracking devices for centimeter-level accuracy and a mixed-integer linear programming model combined with heuristic algorithms to minimize rental and travel costs under operational constraints. Similarly, [Ansola et al. \(2012\)](#) propose a decision support system (DSS) for resource allocation in airports, incorporating multi-agent systems (MAS) and wireless sensor networks. By leveraging RFID readers for real-time resource monitoring, the DSS integrates artificial intelligence with visibility technologies to adapt resource allocation amidst disturbances. Expanding on MAS capabilities, [Ansola et al. \(2013\)](#) develop a distributed MAS-based framework for scheduling ground operations. This approach uses RFID-based feedback to improve dynamic decision making. These studies demonstrate the potential for advanced tracking, scheduling algorithms, and MAS to transform GSE management and airport operations. [Table 6](#) summarizes the literature on GSE handling, categorizing related problems and corresponding optimization techniques.

Several shared modeling assumptions in literature constrain the applicability of the mentioned methodological variety in real airport environments. Most studies formulate GSE scheduling as deterministic optimization problems, where flight schedules, travel speeds, and service durations are treated as fixed inputs without stochastic variability ([Guo et al., 2020](#); [S. Wang et al., 2020](#); [Fink et al., 2019](#); [Andreatta et al., 2014](#); [Padrón et al., 2016](#)). This simplification leads to static time windows derived directly from turnaround plans, which remain unchanged even though real operations frequently shift due to delays, gate swaps, and evolving turnaround process stages. Task structures are also modeled as largely independent chains linked only by precedence

or time-window feasibility ([Fink et al., 2019](#); [Liu et al., 2022](#); [Padrón et al., 2016](#)), thereby neglecting delay propagation, apron congestion, or correlated disruptions across aircraft and resources. Further, while some papers consider coordination between selected vehicle types, joint integration of multiple GSE fleets, staff availability, and workload coupling is usually limited ([Zhao et al., 2019](#); [Liu et al., 2022](#); [Padrón et al., 2016](#)). Additional operational factors such as equipment failure rates, charging/refueling constraints for electrified fleets, and variable handling productivity are generally absent, as vehicles are assumed to be fully reliable with constant speed, capacity, and fuel consumption parameters ([S. Wang et al., 2020](#); [Tabares et al., 2021](#); [Zhao et al., 2021](#); [Andreatta et al., 2014](#)). Collectively, these assumptions, deterministic service times, independent task chains, and static temporal constraints can produce schedules that are theoretically optimal yet sensitive to disruptions and difficult to execute robustly in practice.

5.6. Baggage handling

Baggage handling refers to the process of transporting, sorting, and delivering passengers' luggage from check-in to the aircraft and from the aircraft to baggage claim. Baggage handling comprises three main types of baggage: inbound, transfer, and outbound, where inbound handling manages bags from arriving passengers, transfer handling deals with bags from transit passengers, and outbound handling consolidates bags from transfer and local check-in for delivery to departing flights. However, inefficiencies in planning, system operations, and resource allocation in these processes highlight the need for optimized solutions to enhance overall performance.

The literature on baggage handling focuses on optimizing inbound, outbound and transfer processes to improve efficiency and service quality, especially in resource-constrained environments. For inbound baggage, [Frey et al. \(2017a\)](#) propose a hybrid heuristic that reduces baggage peaks, while [Abdullah et al. \(2024\)](#) optimize luggage offloading with limited ground handling agents. Outbound baggage handling is addressed by [Frey et al. \(2017b\)](#), who minimize workload peaks to avoid disruptions using a time-indexed mathematical programming formulation and column generation. [Ruf et al. \(2022\)](#) develop a rolling planning approach for dynamic baggage loading that adapts to flight delays and passenger uncertainties. Further optimization efforts are made in routing baggage totes to prevent deadlocks. [Sørensen et al. \(2020\)](#) demonstrate that a reinforcement learning agent can effectively manage deadlocks compared to static and dynamic shortest path methods. Additionally, the chute assignment problem in outbound baggage handling, under uncertain conditions, is modeled as a stochastic vector assignment problem by [Huang et al. \(2016\)](#), who use a progressive hedging method. [Huang et al. \(2018\)](#) examine a robust flight unloading area assignment, considering uncertainties in flight delays. These studies collectively examine the key challenges of baggage handling, focusing on minimizing disruptions, improving efficiency, and incorporating uncertainties in both inbound and outbound processes.

Apart from inbound and outbound baggage handling, some authors also specifically focus on the transfer baggage problem. This problem aims to efficiently assign arriving bags to feed areas, from which conveyor belts transport them to outbound flights. [Barth et al. \(2021\)](#) study this problem, with the primary goal of minimizing missed bags, and secondary goals of reducing transportation time, ensuring system robustness, and avoiding overload of handling facilities, using a semi-stochastic model that is solved using a commercial solver.

Another area of focus is the baggage belt assignment problem, which involves assigning arriving flights to baggage belts in the baggage reclaim area of an airport. [Pisinger and Scatamacchia \(2021\)](#) construct a robust schedule for the problem and solved this using dynamic programming. Furthermore, [Barth and Pisinger \(2021\)](#) study the efficient allocation of baggage belts to arriving flights, minimizing passenger wait times while addressing operational requirements. This was modeled as a MIP and solved using a commercial optimization

Table 6
Summary of optimization techniques in ground support equipment handling.

Problem/Focus area	Solution methods	Paper
Special purpose vehicles scheduling optimization	Exact	Han et al. (2020), Zhao et al. (2021)
	Heuristics	Zhao et al. (2019), Guo et al. (2020), Tabares et al. (2021)
	Regression and statistical methods	Ding et al. (2021)
Multi-type vehicles collaborative scheduling optimization	Exact	Andreatta et al. (2016), Fink et al. (2019)
	Heuristics	Andreatta et al. (2014), Padrón et al. (2016), Liu et al. (2022), Zhu et al. (2022), Zhao et al. (2023)
	Simulation based optimization	Liu et al. (2024)
	Learning based approach	Zhou et al. (2023), Y. Wu et al. (2023)
Electric ground support equipment usage	Heuristics	Bao et al. (2023)
	Learning based approach	Ajayi et al. (2023)
	Decision analysis	Sari et al. (2021), Saria et al. (2022)
Optimization of other resources	Heuristics	S. Wang et al. (2020)
	Learning based approach	Ansola et al. (2013, 2012)

solver. J. Wang et al. (2020) address the lack of dynamic, data-driven resource allocation strategies in managing baggage belt assignments, proposing a reinforcement learning-based solution to mine task patterns and incrementally learn effective strategies. Table 7 summarizes the literature on baggage handling, categorizing the related problems and the corresponding optimization techniques.

The studies on baggage handling reveal several common underlying modeling assumptions that may limit real-world applicability. Many works adopt deterministic or scenario-based service times, abstracting away operational variability at carousels, belts, and unloading areas (Huang et al., 2016; Frey et al., 2017a; Pisinger and Scatamacchia, 2021; Ruf et al., 2022). These assumptions simplify baggage arrival profiles, processing rates, and capacity constraints, potentially underestimating congestion and peak-time demands. Further, most formulations treat baggage flows or flight-level workloads as independent task structures, linked only through shared infrastructure capacity without capturing deeper operational couplings such as shared workforce, routing interference, ramp constraints, and correlations in delays (Huang et al., 2016; Frey et al., 2017a,b; Pisinger and Scatamacchia, 2021; Barth and Pisinger, 2021). Baggage handling time is typically managed via static time windows based on scheduled or scenario-based times rather than dynamically updated with real-time delay information (Huang et al., 2016; Barth and Pisinger, 2021; Frey et al., 2017b). While rolling-horizon models introduce some adaptivity, they still rely on deterministic capacity assumptions and may suffer from short-sighted planning (Pisinger and Scatamacchia, 2021; Barth et al., 2021). Even data-driven and learning-based studies operate within simplified simulators that assume stationary dynamics, stylized routing networks, and aggregated state representations (J. Wang et al., 2020; Sørensen et al., 2020). Taken together, these assumptions indicate that existing models may lack robustness when dealing with intrinsic uncertainty, correlated disruptions, and real-time decision dynamics that characterize actual airport baggage handling operations.

5.7. Ground workforce planning and scheduling

Workforce planning involves determining work schedules to meet demand. Workforce scheduling assigns employees to particular workdays and shifts, to meet operational requirements. Leveraging the

available workforce efficiently can yield substantial cost savings. Additionally, effective workforce planning promotes high employee satisfaction, which subsequently enhances productivity. Unlike standard scheduling problems, ground operations must also manage specialized staff and equipment. A key distinction in this domain is that staff requirements are driven by flight schedules and passenger arrival patterns, making accurate demand estimation complex. Recent studies go beyond basic scheduling to address workforce rightsizing and shift design.

Numerous studies have investigated the challenge of optimizing the workforce for airport ground operations. There are two sequential steps in optimizing workforce, which are workforce scheduling and task assignment. Workforce scheduling determines the number of employees working in different time periods, while task assignment determines which tasks are done by which employees. Workforce scheduling and routing problems for baggage loading and unloading have been studied (Dall'Olio and Kolisch, 2023), along with determining staff requirements for outbound baggage loading operations (Cavada et al., 2020). Furthermore, research has focused on identifying the minimum workforce mix that satisfies target coverage rates for given demand profiles in airport ground operations. Dall'Olio and Kolisch (2023) use the branch-and-price approach to solve the workforce scheduling and routing problem. These studies represent some of the key aspects that have been explored under the umbrella of workforce optimization in the aviation industry.

As mentioned previously, the task assignment problem is the next sequential step in optimizing workforce. Studies in this area have considered assigning multiskilled agents to a sequence of tasks in check-in counters (Zamorano et al., 2018) and assigning airline customer service agents to tasks considering staff heterogeneity, time required for moves between locations, and lunch and rest-break requirements (Kuo et al., 2014). Both studies have formulated the problem as a MIP and then used branch-and-price and branch-and-bound to solve the problem.

The ground crew rostering problem is a key challenge in workforce planning. It involves generating monthly shift assignments in the form of rosters for all operating personnel while satisfying various legal, operational, human resource, and welfare constraints. Studies in this area have considered, for instance, such as work patterns (Lusby et al., 2012) and heterogeneous workers (Cavada et al., 2023). Lusby et al. (2012) use a heuristic decomposition approach that utilizes column generation and variable fixing to construct efficient rosters

Table 7
Optimization techniques for baggage handling.

Problem/Focus area	Solution methods	Paper
Inbound and outbound baggage handling	Simulation based optimization	Huang et al. (2018), Abdullah et al. (2024)
	Heuristics	Huang et al. (2016), Frey et al. (2017a,b), Ruf et al. (2022)
	Learning based approach	Sørensen et al. (2020)
Transfer baggage handling	Exact	Barth et al. (2021)
Baggage belt assignment	Exact	Pisinger and Scatamacchia (2021)
	Simulation based optimization	Barth and Pisinger (2021)
	Learning based approach	J. Wang et al. (2020)

for a six-month horizon. Furthermore, Cavada et al. (2023) develop a MILP model and generate solutions through a commercial solver for shift assignments while satisfying various constraints relating to legal restrictions, personnel demand, and worker welfare criteria. For an overview on hiring, scheduling, retention, and digitalization of airport staff, readers are referred to the study by Wandelt and Wang (2024). Table 8 summarizes the literature on ground workforce planning and scheduling, categorizing the related problems and the corresponding optimization techniques.

Certain workforce planning studies relevant to stochastic, dynamic, and multi-skill scheduling did not emerge in our screening results, as they did not explicitly align with the keywords in our search query. Nevertheless, we include these studies to provide contextual completeness, as they address workforce-planning mechanisms such as dynamic rostering, uncertainty-aware shift design, and multi-skill staff allocation in the context of airport ground operations. Pouillet and Parmentier (2020) develop a stochastic task-level shift-planning model for Air France that accounts for delay and outsourcing costs and is solved using column generation. The results show that modeling delay propagation improves schedule resilience and reduces operating costs. Similarly, Z. Wu et al. (2023) argue that deterministic demand curves lack robustness in real operational environments and advocate for stochastic shift design with a scenario-based demand representation to mitigate understaffing and reduce costly last-minute responses such as emergency labor or forced overtime. Further, Hur et al. (2019) develop a stochastic shift-and-break scheduling model for airport baggage handling that adjusts break allocations in response to realized demand, illustrating that stochastic optimization with intraday flexibility outperforms deterministic scheduling based solely on expected demand, especially in volatile flight arrival patterns. Beyond uncertainty, workforce planning research also emphasizes the importance of multi-skill structures and hierarchical staffing. Zeng et al. (2019) present a strategic tour-scheduling model in which higher-skilled personnel may cover lower-skill demands, and solve this using a branch-and-price approach to yield efficient workforce mixes that satisfy service-coverage targets. Taken together, these studies underline the importance of integrating uncertainty modeling, multi-skill considerations, and dynamic rostering mechanisms in airport workforce planning, and reinforce that the literature extends beyond static formulations typically used for shift planning.

Although existing studies in workforce planning and scheduling offer valuable algorithmic and objective-based contributions, they rely on several simplifying modeling assumptions. For example, the majority of papers adopt deterministic service times by treating task processing durations, demand patterns, and staffing requirements as fixed inputs, thereby overlooking stochastic variability in operations (Lusby et al., 2012; Cavada et al., 2020; Dall’Olio and Kolisch, 2023; Zamorano et al., 2018). Furthermore, several models treat tasks as independent structures such that interactions between turnarounds, congestion spill-over, or precedence-driven delay propagation are abstracted away through aggregate staffing levels or simplified networks (Cavada et al., 2020; Číleček et al., 2024; Zamorano et al., 2018; Dall’Olio and Kolisch, 2023). Finally, workforce heterogeneity is often simplified; workers are

assumed fully interchangeable, continuously available, or substitutable without productivity loss (Lusby et al., 2012; Cavada et al., 2023; Číleček et al., 2024). These assumptions may collectively lead to risk, operational bottlenecks, and robustness of generated staffing solutions.

5.8. Interrelationship between key functional areas and solution methodologies

In this section we analyze the interrelationships between the seven key functional areas and the optimization methods applied in the corresponding literature. Fig. 9 presents a system-level mapping of these interrelationships. The corresponding optimization methods are grouped into four categories: exact methods, heuristic methods, learning-based methods, and simulation and hybrid methods. Exact methods include mathematical and exact optimization models, while heuristic methods cover heuristics, metaheuristics, rule-based approaches, and decision analysis. Learning-based methods comprise regression and statistical methods, and data-driven learning approaches, whereas simulation and hybrid methods include discrete-event simulation, Monte Carlo and agent-based simulation, simulation-based optimization, and mathematical modeling combined with simulation. The width of each node and flow reflects the number of studies, highlighting methodological preferences across functional areas. Gate assignment problems predominantly solve using heuristic methods due to their large-scale and combinatorial nature, while turnaround and boarding operations favor simulation and hybrid methods to capture high uncertainty and passenger interactions. Aircraft taxiing and ground movement exhibit a relatively high use of exact and heuristic approaches, whereas ground support equipment handling and baggage handling increasingly employ heuristic, learning-based, and simulation-driven methods to address routing complexity and stochastic task dynamics. Ground workforce planning and scheduling remains largely dominated by exact optimization, reflecting its structured, planning-level characteristics.

5.9. Integrated and joint optimization across multiple ground operations

Recent research has shifted from isolated operational optimization toward integrated models that jointly coordinate gate assignment, taxiing, turnaround, and GSE operations, while increasingly incorporating passenger-centric objectives. Several studies jointly optimize gate assignment and taxiway routing using integrated time-space networks, multi-commodity flow formulations, and set-partitioning approaches, consistently outperforming sequential or priority based methods in terms of taxi time, gate utilization, and total passenger delay (Yang and Gao, 2022; Liu et al., 2023; Jiang et al., 2023b). Importantly, joint gate-runway assignment models extend beyond aircraft-centric efficiency by explicitly incorporating passenger-related performance measures, robustness considerations, passenger taxi-delay reduction, and gate reassignment penalties, as well as environmental objectives such as fuel burn, emissions, and noise, typically addressed through multi-objective evolutionary and branch-and-price algorithms (Behrends and Usher, 2017; Weiszer et al., 2015; Jiang et al., 2024; Hu et al., 2024).

Table 8
Optimization techniques for ground workforce planning and scheduling.

Problem/Focus area	Solution methods	Paper
Workforce scheduling problem	Exact	Cířeček et al. (2024), Dall'Olio and Kolisch (2023)
	Heuristics	Cavada et al. (2020)
Task assignment problem	Exact	Kuo et al. (2014)
	Heuristics	Zamorano et al. (2018)
Ground crew rostering problem	Exact	Cavada et al. (2023)
	Heuristics	Lusby et al. (2012)

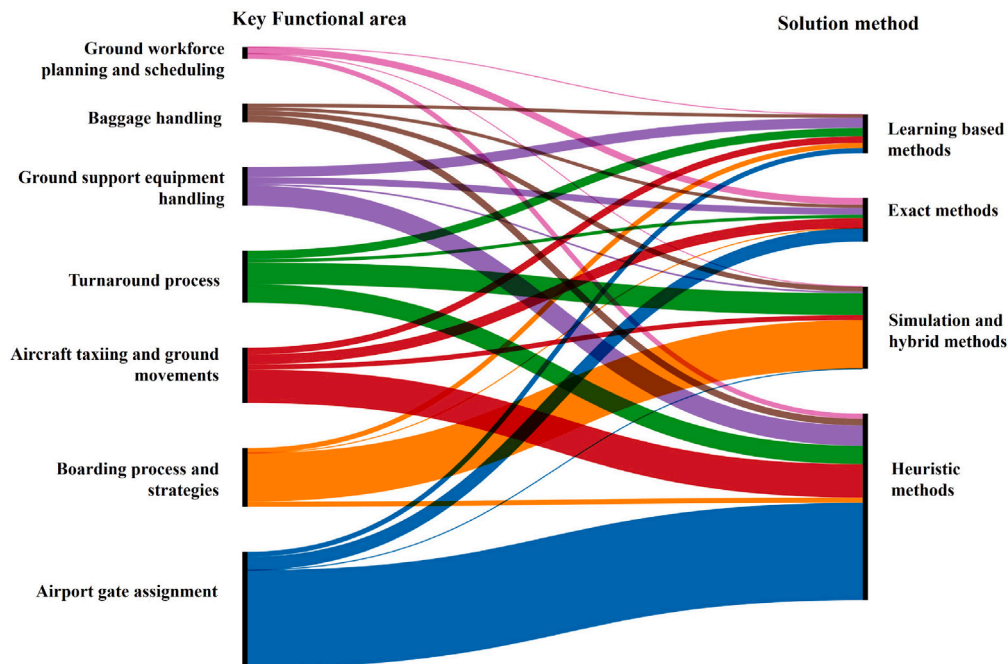


Fig. 9. System-level mapping: interrelationship between 7 key functional areas and optimization methods.

In turnaround operations, integrated recovery and disruption management frameworks combine aircraft assignment, stand allocation, passenger and crew connections, and turnaround resource scheduling, showing that coordinated rebooking, stand reassignment, and accelerated ground handling can fully absorb moderate delay scenarios and reduce total recovery costs (Evler et al., 2022, 2021a,b). Joint optimization models further strengthen strategic-tactical integration by simultaneously addressing fleet assignment, tail assignment, and turnaround handling through decomposition-based mixed-integer programming, thereby capturing critical interdependencies between flight schedules and ground operations (Glomb et al., 2023). At the operational resource layer, joint optimization comes through the coordinated dispatching of GSEs, such as refuellers, shuttle buses, towing tractors, and apron safety fleets. The problems have been addressed through stochastic, chance-constrained, and other advanced optimization models, yielding substantial improvements in service reliability, robustness to arrival time uncertainty, and apron safety (Zhu et al., 2022; Andreatta et al., 2014; Ajayi et al., 2023; Du et al., 2014). Finally, passenger service processes are further strengthened through integrated workforce scheduling for baggage handling, where team formation, skill hierarchies, routing, and task execution are jointly optimized within a branch-and-price-and-check framework, demonstrably outperforming current dispatching policies (Dall'Olio and Kolisch, 2023). Collectively, these examples illustrate a clear evolution toward integrated airport optimization frameworks that explicitly couple infrastructure allocation, vehicle routing, turnaround operations, GSE coordination, and passenger-oriented performance metrics within unified decision-support systems.

5.10. Future research directions

This section outlines the proposed future research directions that emerge from the literature review. Section 5.10.1 highlights specific directions relevant to the specific key functional areas of ground operations. Subsequently, Section 5.10.2 explores broader suggestions categorized into four main themes: algorithmic advancements and computational efficiency, adaptability to uncertainty and real-time conditions, automation and operational synergy, and consideration of other aspects.

5.10.1. Specific future research directions in each key functional area

We outline the future research directions identified through our analysis, segregating them across the seven key functional areas of ground operations.

• Airport gate assignment

Future research should aim to develop models that integrate airline-specific requirements and preferences into the gate assignment problem. These include factors such as gate prioritization, compatibility between gate fleets and airlines, long-term contractual agreements, and collaborative decision-making processes between airlines and airports (Dell'Orco et al., 2017; Kim and Feron, 2017; Deng et al., 2017; Jiang et al., 2013). Additionally, extending the AGAP to address various types of conflicts such as taxiway, runway, and airspace conflicts and their interrelations with gate assignment offers valuable avenues for exploration (Liu et al., 2016). Incorporating measures to support

broader operational and scheduling concerns, such as grouping aircraft into gate clusters, analyzing the availability of support equipment, and dynamically reassigning gates in response to delays, can further enhance operational efficiency (Behrends and Usher, 2017; Maharjan and Matis, 2012). Moreover, integrating other airport operations, including passenger flow management, runway scheduling, baggage handling, and taxi lane sharing, into these models can provide a comprehensive approach to optimizing gate assignments (Kim et al., 2023; She et al., 2022).

- Boarding processes and strategies

Future research on boarding strategies should investigate scenarios with partially filled apron buses and aircraft, compare open seating with faster boarding methods, and empirically evaluate the effects of strategies such as outside-in boarding or skipping half-rows on overall boarding time (Hutter et al., 2019). Promising directions also involve analyzing the optimality and approximation performance of outside-in and back-to-front boarding under conditions of partial bookings or varying congestion, as well as developing constant-factor approximation algorithms (Willamowski and Tillmann, 2022). Investigations should also assess alternative methods by analyzing boarding time sensitivity to unstructured passenger arrivals, variations in occupancy levels, luggage scenarios, and seat assignments (Zeineddine, 2017). Incorporating diverse passenger characteristics such as age, size, gender, and travel experience can significantly enhance the realism of simulation models. For instance, frequent flyers may adopt more efficient behaviors due to greater familiarity with procedures and reduced stress levels (Qiang et al., 2018). Extensions to multiple airplane and boarding door configurations, provide further opportunities for analysis (Delcea et al., 2019). Agent-based models that integrate human behavior are especially promising for improving the realism of boarding simulations (Delcea et al., 2018b). Additionally, boarding strategies for smaller airports with limited jet-bridge facilities should be explored (Nugroho and Asrol, 2022). Developing safety risk models that account for COVID-19 factoring in aircraft type, vulnerable passenger grouping, passenger experience, and strategic seat placement can also contribute to safer and more efficient boarding processes (Salari et al., 2022). Lastly, investigating the reverse pyramid boarding strategy with more than three unequal-size boarding groups, accounting for passengers traveling with or without hand luggage, could provide insights into strategies suited to situations with safety concerns (Delcea et al., 2020).

- Turnaround process

Future research should explore the impact of airport configurations, information system capabilities, and diverse scheduling approaches on aircraft turnaround time and system costs, particularly considering varying arrival patterns of incoming aircraft (Wu and Caves, 2004a). The influence of weather conditions, including visibility at arrival and departure, on actual turnaround times and scheduled operations deserves further investigation (Positorino et al., 2020). Comparing optimization-based solutions with traditional traffic management approaches, while adapting algorithms to dynamic rescheduling environments with rolling horizons and uncertainty, presents another valuable avenue for research (D'Ariano et al., 2012). Additionally, the application of advanced technologies such as RFID sensors and barcode readers to automate turnaround processes could improve efficiency (Abd Allah Makhloof et al., 2014). Exploring the potential of autonomous vehicles and other emerging technologies to enhance precision and reduce human error in aircraft turnaround operations also holds significant promise for future advancements.

- Aircraft taxiing and ground movements

Future research should explore more comprehensive and integrated optimization of runway crossings, considering multiple

runways and gate operations. This could involve using alternative speed profiles for aircraft and testing the approach at airports with different layouts and traffic levels (Guépet et al., 2017). Additionally, research can delve deeper into operations such as holding positions, ground speed variations, variable time separations, and uncertainty in maneuvers (García et al., 2005; Ravizza et al., 2014a). Realistic scenarios specific to aircraft taxiing, including fuel consumption, noise and emissions, pilot behavior, and different aircraft configurations, should also be considered (Weiszer et al., 2020). Lastly, reducing uncertainty in taxi times, alongside managing allocated time slots, priority, and integrating arrival processes, should be a key focus for future studies (Idrissi et al., 2022).

- Ground support equipment handling

Future research could extend the GSE handling problem by addressing the real-time scheduling of ferry vehicles in response to uncertain and unexpected situations (Han et al., 2020). Integrating the ground vehicle assignment and scheduling problem with other optimization problems such as staff scheduling, gate assignment, taxiway planning, and conflict detection and resolution offers another promising research direction (Andreatta et al., 2014). Incorporating data from cameras, infrared devices, and sensors into the airport ground movement system can enhance data collection, facilitate monitoring, and identify ramp risks such as collisions, thereby addressing safety concerns in VRPs (S. Wang et al., 2020). Introducing electric vehicles and charging stations into the system and assessing their impact on vehicle scheduling could provide insights into sustainable practices (Saria et al., 2022). Additionally, extending the vehicle scheduling problem to include unmanned technologies, such as autonomous ferry and fueling trucks, and investigating cooperative optimization techniques that combine manual and automated scheduling presents a valuable area for exploration (Sari et al., 2021).

- Baggage handling

Future research directions in baggage handling systems could include incorporating “destination time frames”, which refer to specific deadlines for baggage to reach destinations, adding complexity to the problem while enhancing its real-world applicability. This also offers an opportunity to explore the scalability and performance of routing solutions (Sørensen et al., 2020). The baggage belt assignment problem could be examined through the lens of the position dependent knapsack problem from both theoretical and practical perspectives, while the online chute assignment problem presents another promising area for study (Pisinger and Scatamacchia, 2021; Huang et al., 2016). Additionally, future work could consider realistic operational factors such as worker capacity, shifts, breaks, and individual worker movements, along with the dynamic rates at which bags are fed into the system and the unit load devices (Ruf et al., 2022).

- Workforce planning and scheduling

Future research on staff planning should incorporate additional features such as employee preferences, synchronized agent arrivals, and the option for agents to leave their stations early (Zamorano et al., 2018). Integrating staff planning into the broader operational workforce planning processes, while accounting for the preferences and skills of handlers in shift and task assignments, represents another important research direction (Cavada et al., 2020). Moreover, workforce planning in the context of baggage handling could benefit from addressing related tasks such as cart transportation, baggage sorting, and security screening, thereby providing a more comprehensive approach to optimizing operations (Cavada et al., 2020). Dall'Olio and Kolisch (2023) emphasize the need for future research on incorporating uncertainty into workforce optimization for the baggage loading and unloading problem. Factors such as equipment failures can impact task durations, underscoring the necessity for more robust approaches in this domain.

5.10.2. A comprehensive exploration of research directions in four key themes

While existing studies propose a wide range of future research opportunities, these directions are often scattered across functional areas and lack integration into a coherent research agenda. To provide a more structured outlook, we organize future research directions into four targeted themes, each addressing a clearly identified methodological or operational gap in airport ground operations. The four main themes are: algorithmic advancements and computational efficiency, adaptability to uncertainty and real-time conditions, automation and operational synergy, and considerations of other aspects.

1. Algorithmic advancements and computational efficiency

Despite significant progress in optimization-based models for airport operations, most existing approaches remain problem-specific and computationally expensive, limiting their scalability to large, integrated airport systems. Building on the reviewed literature, several overarching research directions emerge: hybrid optimization, model enrichment, AI integration, and computational scaling.

Hybrid Optimization: A key limitation of many current approaches is their reliance on a single optimization paradigm, which often leads to trade-offs between solution quality and computational tractability. To address this, multiple studies suggest hybrid frameworks that combine integer programming, constraint programming, and metaheuristics to exploit their complementary strengths (Ornek et al., 2022; Dorndorf et al., 2012; Asadi et al., 2021). Metaheuristic-based extensions using particle swarm optimization and genetic algorithms have also been proposed to enhance solution quality in complex operational problems (Gołda et al., 2019).

Model Enrichment: Another critical gap lies in the limited representation of operational realism in many existing formulations. Future research is encouraged to incorporate richer operational constraints and stakeholder preferences, including airline alliances, and by considering multiple objectives and evaluating the impact of different cost functions (Diepen et al., 2012; Jiang et al., 2023c; Cai et al., 2019; Lim et al., 2005). Extensions toward taxiing optimization based on aircraft size and passenger volume, as well as heterogeneous GSE fleets with synchronized workforce rostering, remain largely unexplored (Padrón et al., 2016).

AI Integration: Beyond classical optimization, AI offers promising capabilities for handling uncertainty, learning system behavior, and enabling real-time control. Neural and Bayesian networks can improve predictive accuracy under demand and delay uncertainty, fuzzy systems can model ambiguous operational conditions, and deep reinforcement learning can support adaptive decision-making in real-time scheduling and disruption recovery (Ansola et al., 2013; Gök et al., 2023; Zhou et al., 2023). These techniques are particularly relevant for highly dynamic processes such as gate reassignment, baggage handling, and workforce reassignment.

Computational Scaling: As airport systems grow in size and connectivity, computational scalability becomes a binding constraint. Several authors emphasize the need for advanced hardware infrastructures, parallel computation schemes, and distributed computing environments to ensure that large-scale, integrated airport models remain solvable within operational time limits, especially for hub airports with dense traffic volumes (Stollenwerk et al., 2020; Yang and Gao, 2022).

2. Adaptability to uncertainty and real-time conditions

Although uncertainty is widely acknowledged in airport operations, most existing models either simplify stochastic behavior or evaluate robustness only through limited scenario testing, highlighting the need for improved treatment of operational

uncertainty, large-scale validation, and robust optimization approaches.

Operational uncertainty: Researchers have emphasized the importance of explicitly modeling the inherent uncertainty and dynamic nature of airport operations. Given the complexity of large-scale airport environments, particularly at major international hubs, advanced methodologies are required to handle fluctuating operational conditions and unpredictable disruptions (Kim et al., 2023; Li et al., 2024). Studies highlight the necessity of developing dynamic models that can adapt to real-time variations in flight schedules, passenger flows, resource availability, and operational constraints.

Large-scale validation: The literature further stresses the need to evaluate optimization models in real-world operational settings and to test them on large-scale or multi-hub airline networks to ensure their robustness and scalability under dynamic conditions (Evler et al., 2022).

Robust optimization: Robustness in airport operations can be addressed using stochastic optimization, simulation-based optimization, and emerging approaches such as quantum heuristics. Several studies emphasize the need to develop more advanced robust methods capable of generating reliable and cost-effective schedules under uncertainty (Guépet et al., 2015; Dorndorf et al., 2007; Silva et al., 2023; Stollenwerk et al., 2020).

3. Automation and operational synergy

Current automation-focused studies primarily address isolated decision problems, while practical airport operations require tightly coupled coordination across gates, runways, ground vehicles, and terminal processes, motivating the need for enhanced sensor integration and more comprehensive resource coupling.

Sensor Integration: Further research is needed to more effectively embed automation technologies into ground operations while simultaneously reducing computational time. Several studies propose the use of Internet of Things (IoT) systems and video analytics to enable real-time monitoring of aircraft movements, gate occupancy, and passenger flows, thereby strengthening data-driven decision-making for gate assignment and terminal operations (Karsu and Solyali, 2023).

Resource Coupling: Other studies emphasize the importance of integrating traditionally independent airport decision-making problems, such as stand allocation, taxiway routing, runway scheduling, and ground service vehicle dispatching, into unified optimization frameworks (Guépet et al., 2015; Zeunert and Herrich, 2022; Y. Li et al., 2021). Several of these integrations have already been successfully realized in the literature, as discussed in Section 5.9, where joint optimization of gates, taxiing, turnaround processes, GSE operations, and workforce scheduling demonstrates clear system-wide benefits over sequential approaches. As highlighted by Asadi et al. (2021), boarding strategies could be further optimized through adaptive group sizing, while turnaround models should integrate aircraft stand allocation, GSE coordination, and shared resource management (Gök et al., 2023). Such tightly coupled formulations are essential for capturing operational interdependencies and for achieving system-wide performance improvements rather than localized optimizations.

4. Consideration of other aspects

Health safety, passenger experience, stakeholder interactions, decision transparency and environmental sustainability remain underrepresented in most ground operation models, which still prioritize operational efficiency as the dominant performance metric.

Health safety: Passenger safety, especially mitigation of contamination risks, has become a critical concern in ground operations, especially following the COVID-19 pandemic. In response,

various measures have been implemented to enhance safety during boarding processes (Qureshi and Qureshi, 2022).

Passenger experience: Given the importance of customer experience for airlines, investigating the impact of ground operations on passenger satisfaction could provide valuable insights (Liang et al., 2020; Zhang et al., 2022). To ensure unbiased feedback, conducting interviews with passengers immediately after key processes such as boarding, gate assignments, and baggage handling would be beneficial (Bachmat et al., 2023).

Stakeholder coordination: Another aspect is the interaction between stakeholders, particularly in resource allocation, which can be optimized to reduce delays and disruptions (Chen et al., 2016). To achieve this, studies should focus on improving coordination among entities involved in ground operations, ensuring more efficient operations and reducing disruptions (Gładys et al., 2022). Research could also explore the fairness of the sequencing process by developing assessment methods to evaluate how well the system serves all stakeholders and identifying approaches to enhance equity in flight scheduling (Karapetyan et al., 2017).

Decision transparency: Enhancing the transparency and comprehensibility of the system for airport operators would allow them to readily override decisions when necessary, further improving flexibility and responsiveness (Karapetyan et al., 2017; Cavada et al., 2023).

Environmental sustainability: Finally, optimization models can be extended to incorporate multi-objective goals, such as minimizing emissions, noise, and fuel consumption, contributing to more sustainable airport operations (Deng et al., 2017; Weiszer et al., 2020; Silva et al., 2023).

6. Conclusion

This SLR has provided an extensive overview of airport ground operations and commonly used methods to optimize these processes, highlighting various critical functional areas such as gate assignment, boarding strategies, turnaround processes, aircraft taxiing, ground support equipment handling, baggage handling, and workforce planning and scheduling. By analyzing 199 research papers published over the past two decades, we have mapped the diverse optimization problems, methodologies, and advancements in the field. The findings underscore the vital role of operations research techniques, particularly mathematical modeling, metaheuristics, and simulation-based methods, in enhancing the efficiency of airport ground operations. The purpose of the present study was not to cover all existing literature on the topic, but rather to analyze a significant sample size to provide an overview to researchers and practitioners on the developments in the field and to guide future developments in this area.

Our review highlights substantial advancements in the optimization of airport ground operations, particularly in the areas of airport gate assignment and aircraft turnaround processes, which have received the most attention from researchers. However, areas such as baggage handling and ground workforce planning and scheduling remain under-explored, presenting valuable opportunities for further investigation. While a variety of optimization techniques, including mixed integer programming, metaheuristics, and simulation models, have been successfully applied, the integration of real-time data, adaptability to disruptions, and development of robust solutions to address uncertainties have emerged as crucial areas requiring additional research. Future research should focus on inclusion of stochasticity in the decision-making processes and the automation of airport operations to enhance operational resilience. Additionally, advancements in computational techniques, such as machine learning, digital twins, and reinforcement learning, are anticipated to revolutionize airport ground operations, as they also do in other operations research domains. The optimization of airport ground operations is a rapidly advancing field that holds significant importance for the aviation industry. This comprehensive

review not only summarizes the current research landscape but also highlights crucial areas for future exploration, laying the groundwork for improving the efficiency, safety, and sustainability of airport ground operations.

CRedit authorship contribution statement

Mahekha Dahanayaka: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Dennis Prak:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Martijn Mes:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

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Data availability

Data will be made available on request.

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